

COMMUTATIVE ALGEBRA - FALL 2024

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1. RINGS AND MODULES

1.1. Commutative rings and ideals. We begin by recalling some basic definitions and facts in commutative algebra.

Definition 1.1.

- A **commutative ring**¹ is a set R with two operations $+$ and $\cdot$ so that
 - (1) $(R, +)$ is an abelian group with identity 0 ;
 - (2) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$, for any $a, b, c \in R$;
 - (3) $a \cdot (b + c) = a \cdot b + a \cdot c$, for any $a, b, c \in R$;
 - (4) $a \cdot b = b \cdot a$, for any $a, b \in R$;
 - (5) $a \cdot 1 = 1 \cdot a = a$, for any $a \in R$.
- Let R be a commutative ring. An element $a \in R$ is called a **unit** if $a \neq 0$ and a is invertible, i.e. there exists $b \in R$ such that $a \cdot b = 1$. If this holds, then b is called the **inverse** of a .
- A **field** is a commutative ring R such that $(R \setminus \{0\}, \cdot)$ is also a group. Equivalently, R is a field if each non-zero element of R is a unit.
- Let R be a commutative ring. Then $a \in R$ is called a **zero divisor** if $a \neq 0$ and there exists $b \in R$ such that $a \cdot b = 0$. A commutative ring R is called an **integral domain** if it does not admit any zero divisors.
- Let R_1 and R_2 be commutative rings. A function $f: R_1 \rightarrow R_2$ is called a **morphism of rings** (or **homomorphism**) if
 - (1) $f(a + b) = f(a) + f(b)$, for any $a, b \in R_1$;
 - (2) $f(a \cdot b) = f(a) \cdot f(b)$, for any $a, b \in R_1$;
 - (3) $f(1) = 1$.
 Moreover, f is called an **isomorphism** of rings if it is bijective.

We will often just refer to morphisms of rings simply as morphisms, when it is clear from the context.

Example 1.2. Let m be a positive integer. We denote by

$$\mathbb{Z}/m\mathbb{Z} := \{[0], [1], \dots, [m-1]\}$$

the set of numbers modulo m , where

$$[a] = \{a + km \mid k \in \mathbb{Z}\}.$$

Then $\mathbb{Z}/m\mathbb{Z}$ is a commutative ring and the following are equivalent

- (1) $\mathbb{Z}/m\mathbb{Z}$ is a field;
- (2) $\mathbb{Z}/m\mathbb{Z}$ is an integral domain;
- (3) m is prime.

In general, the units of $\mathbb{Z}/m\mathbb{Z}$ are the elements $[a] \in \mathbb{Z}/m\mathbb{Z}$ such that $1 \leq a \leq m-1$ and $\gcd(a, m) = 1$. The zero divisors are the elements $[a] \in \mathbb{Z}/m\mathbb{Z}$ such that $1 \leq a \leq m-1$ and a divides m .

¹Note that we will always assume that our rings admit an identity with respect to the product. In the literature, these might be referred to as "commutative rings with identity"

Example 1.3. Let k be a field. We denote $R = k[x]$ to be the set of polynomials in one variable x and with coefficients in k , i.e.

$$R = \{a_0 + a_1x + \cdots + a_mx^m \mid a_0, \dots, a_m \in k\}.$$

More in general we denote by $k[x_1, \dots, x_n]$ the ring of polynomials in variables $x_1, \dots, x_n$. Then R is an integral domain. The units of R are all the non-zero elements of k .

Recall that the field k is **algebraically closed** if for any non-constant polynomial $f \in k[x]$ there exists $a \in k$ such that $f(a) = 0$.

Definition 1.4. Let R be a commutative ring and let $S \subset R$ be a subset. Then S is called **multiplicative** if $1 \in S$ and for any $a, b \in S$ we have that $a \cdot b \in S$.

Proposition 1.5. Let R be an integral domain. Then there exists a set

$$\text{Frac}(R) := \left\{ \frac{a}{b} \mid a, b \in R, b \neq 0 \right\}$$

which admits two natural operations so that it is a field which contains R as a subring. We will refer to $\text{Frac}(R)$ as the **field of fractions** of R .

Proof. This is an easy exercise. □

More in general we have:

Proposition 1.6. Let R be a commutative ring and let $S \subset R$ be a multiplicative ring. Define the equivalence $\sim$ on $R \times S$ so that

$$(a, s) \sim (b, s') \quad \text{if and only if} \quad \exists t \in S \text{ such that } t(as' - bs) = 0$$

and denote by $\frac{a}{s}$ the equivalence class of (a, s) . Then the set

$$S^{-1}R := \left\{ \frac{a}{s} \mid a \in R, s \in S \right\}$$

admits two natural operations so that it is a commutative ring². If R is an integral domain then there is a natural inclusion $R \hookrightarrow S^{-1}R$. We will refer to $S^{-1}R$ as the **ring of fractions** (or **localisation**) of R with respect to S .

Proof. This is also an easy exercise. □

Definition 1.7. Let R be a commutative ring. Then $a \in R$ is called **nilpotent** if there exists a positive integer n such that $a^n = 0$.

Example 1.8. $[2] \in \mathbb{Z}/8\mathbb{Z}$ is nilpotent.

Exercise 1.9. Let R be a commutative ring and let $a \in R$ be a non-zero element. Define

$$S = \{a^n \mid n \geq 0\}.$$

Then S is multiplicative and we may denote

$$R_a := S^{-1}R.$$

Show that $R_a = \{0\}$ if and only if a is nilpotent.

²Note that $S^{-1}R$ might not admit a unit (cf. Exercise 1.9)

Proposition 1.10. *Let R be a commutative ring and let $a \in R$ be nilpotent. Then $1 + a$ is a unit.*

Proof. Let n be a positive integer such that $a^n = 0$. Then it is easy to check that

$$\sum_{i=0}^{n-1} (-a)^i = 1 - a + a^2 + \cdots + (-a)^{n-1}$$

is the inverse of $1 + a$. □

Definition 1.11. *Let R be a commutative ring. An **ideal** I of R is a subset $I \subset R$ such that*

- (1) I is a subgroup of $(R, +)$;
- (2) for any $r \in R$ and $a \in I$, we have $ra \in I$.

An ideal I of R is called **proper** if $I \neq R$.

Proposition 1.12. *If R is a commutative ring and $I \subset R$ is an ideal, then R/I is also a commutative ring with a natural morphism of rings*

$$R \rightarrow R/I.$$

Proof. This is an easy exercise. □

Example 1.13. *If $R = \mathbb{Z}$ and $I = m\mathbb{Z} = \{mz \mid z \in \mathbb{Z}\}$ then I is an ideal of R and $R/I = \mathbb{Z}/m\mathbb{Z}$ as in example 1.2.*

Definition 1.14. *Let R be a commutative ring and let $S \subset R$ be a subset. The **ideal generated** by S is*

$$(S) := \{r_1 s_1 + \cdots + r_n s_n \mid r_1, \dots, r_n \in R, s_1, \dots, s_n \in S\}.$$

If $S = \{s_1, \dots, s_q\}$ is a finite set, then we denote

$$(s_1, \dots, s_q) := (S).$$

Exercise 1.15. *Let R be a commutative ring.*

- (1) *Let $S \subset R$ be a subset. Check that (S) is an ideal of R .*
- (2) *Show that $(1) = R$.*
- (3) *Let $x \in R$. Show that x is a unit if and only if $(x) = R$.*

Definition 1.16. *Let R be a commutative ring. An ideal I is called **principal** if there exists $x \in R$ such that $I = (x)$. A ring is called a **principal ideal domain** (or **PID**) if all its ideals are principal.*

Example 1.17. *Recall that an integral domain R is called an **Euclidean domain** if there exists a norm*

$$|\cdot|: R \rightarrow \mathbb{N}$$

such that, for any $a, b \in R$, with $b \neq 0$, there exist $q, r \in R$ such that

$$a = bq + r \quad \text{and} \quad |r| < |b|.$$

It is an easy exercise to show that any Euclidean domain is a PID. In particular, $\mathbb{Z}$ is a PID. Moreover, if k is a field then $R = k[x]$ is a PID.

Example 1.18. Recall that an integral domain R is called a **unique factorisation domain** (or **UFD**) if for any $a \in R$, there exist irreducible elements $p_1, \dots, p_m \in R$ such that

$$a = p_1 \cdots p_m$$

Moreover such a decomposition is unique up to the choice of the order $p_1, \dots, p_m$ and multiplication by units.

We have that any PID is a UFD.

Exercise 1.19. Show that

$$\mathbb{Z}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$$

is a PID.

Definition 1.20. Let R be a commutative ring and let $I, J \subset R$ be ideals. We denote

$$I + J := \{a + b \mid a \in I, b \in J\} \subset R$$

and

$$I \cdot J := \{a_1 b_1 + \cdots + a_n b_n \mid a_i \in I, b_i \in J\} \subset R.$$

In particular, if $I = (x)$ for some $x \in R$ then we denote

$$xJ := (x) \cdot J \subset R.$$

The **quotient ideal** is

$$(I : J) := \{x \in R \mid xJ \subset I\}.$$

Finally, if $k \geq 1$ then the **k -th power** of I is the ideal I^k generated by the set

$$S = \{r_1 \cdots r_k \mid r_1, \dots, r_k \in I\}.$$

The ideal $I \subset R$ is called **nilpotent** if there exists a positive integer k such that $I^k = (0)$.

Exercise 1.21. Let R be a commutative ring and let $I, J \subset R$ be ideals.

- (1) $I + J$, $I \cdot J$ and $(I : J)$ are also ideals of R .
- (2) if $J \subset I$ then $(I : J) = R$;
- (3) $(I : R) = I$;
- (4) $(R : I) = R$.

1.2. Prime and maximal ideals.

Definition 1.22. Let R be a commutative ring and let $\mathfrak{m} \subset R$ be a proper ideal. Then $\mathfrak{m}$ is called **maximal ideal** if for any ideal $I \subset R$ such that $\mathfrak{m} \subset I$, we have that $\mathfrak{m} = I$ or $\mathfrak{m} = R$.

Exercise 1.23. Let R be a commutative ring and let $\mathfrak{m} \subset R$ be a proper ideal. Show that $\mathfrak{m}$ is a maximal ideal if and only if $R/\mathfrak{m}$ is a field.

Example 1.24. Let $R = \mathbb{Z}$, let $m \in R$ be a non-zero number and let $I = (m)$. Then, as in Example 1.2, we have that $\mathbb{Z}/I = \mathbb{Z}/m\mathbb{Z}$ is a field if and only if m is prime. Thus, I is a maximal ideal if and only if m is prime.

Example 1.25. Let k be a field and let $R = k[x]$. Let $a \in k$ and let $\mathfrak{m} = (x - a) \subset R$ is a maximal ideal. Indeed, the function

$$R/\mathfrak{m} \rightarrow k \quad [f(x)] \mapsto f(a)$$

is a well-defined isomorphism. Thus, $R/\mathfrak{m}$ is a field and, by the previous exercise, it follows that $\mathfrak{m}$ is maximal.

More in general, if $R = k[x_1, \dots, x_m]$ and $a_1, \dots, a_m \in k$ then

$$\mathfrak{m} := (x - a_1, \dots, x - a_m) \subset R$$

is a maximal ideal.

We will show that if k is an algebraically closed field then these are the only maximal ideals (cf. Corollary 3.21). On the other hand, if $k = \mathbb{R}$ then $\mathfrak{m} := (x^2 + 1) \subset \mathbb{R}[x]$ is maximal, even though it is not of the form $(x - a_1, \dots, x - a_m)$. Note that if $I := (x^2 - 1) \subset \mathbb{R}[x]$ then I is not maximal as $I \subset (x - 1)$.

Definition 1.26. Let P be a set. Then

- P is **partially ordered** if it admits a relation $\leq$ such that
 - (1) $a \leq a$ for any $a \in P$;
 - (2) if $a, b \in P$ are such that $a \leq b$ and $b \leq a$ then $a = b$;
 - (3) if $a, b, c \in P$ are such that $a \leq b$ and $b \leq c$ then $a \leq c$.
- P is **totally ordered** if it is partially ordered and for any $a, b \in P$, we have $a \leq b$ or $b \leq a$.
- Assume that P is partially ordered and $T \subset P$ is a subset. Then an **upper bound** of T is an element $x \in P$ such that $a \leq x$ for all $a \in T$.
- Assume that P is partially ordered and $m \in P$. Then m is called **maximal** if for any $a \in P$ such that $m \leq a$, we have $a = m$.

The following result is equivalent to the Axiom of Choice:

Theorem 1.27 (Zorn's Lemma). Let P be a partially ordered set such that every totally ordered subset $T \subset P$ has an upper bound. Then P admits a maximal element $m \in P$.

Proposition 1.28. Let R be a commutative ring and let $I \subset R$ be a proper ideal. Then there exists a maximal ideal $\mathfrak{m} \subset R$ containing I .

Proof. Let P be the set of proper ideals in R containing I . Then we can order the elements of P by inclusion. Let $T \subset P$ be a totally ordered subset. Then it is easy to check that

$$I := \bigcup_{J \in T} J \subset R$$

is a proper ideal of R . Thus, I is an element in R which is an upper bound of T . By Zorn's Lemma, P admits a maximal element, which is a maximal ideal of R . $\square$

Definition 1.29. Let R be a commutative ring and let $\mathfrak{p} \subset R$ be a proper ideal. Then $\mathfrak{p}$ is said to be **prime** if for any $x, y \in R$ such that $x \cdot y \in \mathfrak{p}$, we have that $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$.

Exercise 1.30. Let R be a commutative ring and let $I \subset R$ be a proper ideal. Show that

- (1) I is prime if and only if R/I is an integral domain;
- (2) every maximal ideal is prime.

Exercise 1.31. Let $f: R_1 \rightarrow R_2$ be a morphism of rings and let $I \subset R_2$ be an ideal.

- (1) Show that if I is prime then $f^{-1}(I)$ is also prime.
- (2) Show that if I is maximal then $f^{-1}(I)$ is not necessarily maximal.

Example 1.32. As in Example 1.24, let $R = \mathbb{Z}$ and let $m \in R$ be a non-zero element. Then the following are equivalent:

- (1) $(m) \subset R$ is prime;
- (2) $(m) \subset R$ is maximal;
- (3) m is prime.

Example 1.33. Let k be a field and let $R = k[x, y]$. Let $I = (x) \subset R$. Then the function

$$R/I \rightarrow k[y] \quad [f(x, y)] \mapsto f(0, y)$$

is an isomorphism. As in Example 1.3, we have that $k[y]$ is an integral domain but not a field. Thus, I is prime in R but not maximal.

On the other hand, the ideals $(x^2) \subset R$ and $(xy) \subset R$ are not prime.

Proposition 1.34. Let R be a commutative ring and let $I \subset R$ be an ideal. Denote

$$\sqrt{I} := \{r \in R \mid r^n \in I \text{ for some } n > 0\}.$$

Then $\sqrt{I} \subset R$ is an ideal. We will refer to $\sqrt{I}$ as the **radical** of I .

Proof. The most difficult part is to show that if $x, y \in \sqrt{I}$ then $x + y \in \sqrt{I}$. Let n and m be positive integers such that $x^n, y^m \in I$. By the binomial formula, we have that

$$(x + y)^{n+m-1} = \sum_{i=0}^{n+m-1} r_i x^i y^{n+m-1-i}$$

for some $r_i \in R$. Note that for any $i = 0, \dots, n + m - 1$, we have that either $i \geq n$ or $n + m - 1 - i \geq m$. In both cases, we have that $x^i y^{n+m-1-i} \in I$ and therefore $(x + y)^{n+m-1} \in I$. Thus, $x + y \in \sqrt{I}$. $\square$

Exercise 1.35. Let R be a commutative ring. Show that the following are equivalent:

- (1) R has exactly one prime ideal;
- (2) every element of R is either a unit or nilpotent;
- (3) $\sqrt{(0)}$ is a maximal ideal.

1.3. Geometric point of view. Let k be a field and let n be a positive integer. Then we denote

$$\mathbb{A}_k^n := \{(c_1, \dots, c_n) \mid c_i \in k \text{ for } i = 1, \dots, n\}$$

the n -**dimensional affine space** over k ³. Let $S \subset k[x_1, \dots, x_n]$ be a set of polynomials. Then the zero locus

$$V(S) := \{x \in \mathbb{A}_k^n \mid f(x) = 0 \text{ for all } f \in S\} \subset \mathbb{A}_k^n$$

is called an **affine variety** (or simply **variety**). If $S = \{f_1, \dots, f_k\}$ is finite, then we just denote $V(S)$ as $V(f_1, \dots, f_k)$. Note that if $S_1 \subset S_2 \subset k[x_1, \dots, x_n]$ then $V(S_2) \subset V(S_1)$. If $I := (S) \subset k[x_1, \dots, x_n]$ is the ideal generated by a subset $S \subset k[x_1, \dots, x_n]$ then $V(S) = V(I)$. Thus, we can always assume that a variety is defined by an ideal I of $k[x_1, \dots, x_n]$.

If $V, W \subset \mathbb{A}_k^n$ are varieties such that $W \subset V$ then W is called a **subvariety** of V .

Example 1.36. If we take $S = \{0\} \subset k[x_1, \dots, x_n]$ then $V(S) = \mathbb{A}_k^n$. On the other hand, if $S = k[x_1, \dots, x_n]$ then $V(S) = \emptyset$. Thus, $\mathbb{A}_k^n$ and $\emptyset$ are affine varieties.

Lemma 1.37. Let k be a field and let $I, J \subset k[x_1, \dots, x_n]$ be ideals and let $V = V(I) \subset \mathbb{A}_k^n$ and $W = V(J) \subset \mathbb{A}_k^n$ be the associated varieties. Then

$$V \cap W = V(I + J)$$

and

$$V \cup W = V(I \cdot J).$$

In particular, the intersection and the union of V and W are also varieties.

Proof. This is an easy exercise. □

Let k be a field. We now assume that $X \subset \mathbb{A}_k^n$ is a subset, and we define

$$I(X) := \{f \in k[x_1, \dots, x_n] \mid f(x) = 0 \text{ for all } x \in X\}.$$

It is easy to check that $I(X) \subset k[x_1, \dots, x_n]$ is an ideal, which we will call the **ideal** of X . Note that if $X_1 \subset X_2 \subset \mathbb{A}_k^n$ then $I(X_2) \subset I(X_1)$.

Lemma 1.38. Let k be a field and let $X \subset \mathbb{A}_k^n$ be a subset. Then $V(I(X))$ is the smallest variety in $\mathbb{A}_k^n$ which contains X .

In particular, if X is a variety, then $V(I(X)) = X$.

Proof. Let $x \in X$. Then $f(x) = 0$ for all $f \in I(X)$ and, therefore, $x \in V(I(X))$. Thus,

$$X \subset V(I(X)).$$

Assume now that $W \subset \mathbb{A}_k^n$ is a variety such that $X \subset W$. We want to show that $V(I(X)) \subset W$. Suppose not. Let $J \subset k[x_1, \dots, x_n]$ be an ideal such that $W = V(J)$. Then J is not contained in $I(X)$ and there exists $f \in J$ which is not contained in

³Note that as a set $\mathbb{A}_k^n$ coincides with k^n , but it is common to distinguish the notation when we refer to the affine space.

$I(X)$. Thus, there exists $x \in X$ such that $f(x) \neq 0$, which implies that $x \notin W$, a contradiction. $\square$

Lemma 1.39. *Let k be a field and let $I \subset k[x_1, \dots, x_n]$ be an ideal. Then*

- (1) $V(I) = V(\sqrt{I})$; and
- (2) $\sqrt{I} \subset I(V(I))$.

Proof. This is an easy exercise. $\square$

We will show that if k is algebraically closed and $I \subset k[x_1, \dots, x_n]$ is an ideal then

$$I(V(I)) = \sqrt{I}$$

(cf. Theorem 3.23).

Given $X \subset \mathbb{A}_k^n$, we define

$$A(X) := k[x_1, \dots, x_n]/I(X)$$

to be the **coordinate ring** of X . Note that any $[f] \in A(X)$ defines a function

$$\tilde{f}: X \rightarrow k \quad x \mapsto f(x).$$

It is easy to check that $\tilde{f}$ is well defined.

Assume now that $X \subset \mathbb{A}_k^n$ and $Y \subset \mathbb{A}_k^m$. Let

$$F: \mathbb{A}_k^n \rightarrow \mathbb{A}_k^m \quad F = (f_1, \dots, f_m)$$

where $f_1, \dots, f_m \in k[x_1, \dots, x_n]$ and such that $F(X) \subset Y$. The restricted function

$$\varphi := F|_X: X \rightarrow Y$$

is called a **regular mapping** or a **morphism between varieties**.

Note that we have an induced morphism of rings

$$A(Y) \rightarrow A(X) \quad [g] \mapsto [g \circ F] = [g(f_1, \dots, f_m)].$$

It is again easy to check that this morphism is well defined.

Example 1.40. *Let k be a field and let $a := (a_1, \dots, a_n) \in \mathbb{A}_k^n$ be a point. Then if $X := \{a\}$, the ideal of X is*

$$I(X) = (x_1 - a_1, \dots, x_n - a_n)$$

is the maximal ideal as in Example 1.25.

Definition 1.41. *Let k be a field and let $I \subset k[x_1, \dots, x_n]$ be an ideal and let $V := V(I)$. Then V is said to be **irreducible** if V cannot be written as a union $V = V_1 \cup V_2$ where $V_i \subset V$ is a proper affine subvariety for $i = 1, 2$.*

Lemma 1.42. *Let k be a field and let $V \subset \mathbb{A}_k^n$ be an affine variety. Then V is irreducible if and only if $I(V)$ is a prime ideal.*

Proof. Assume first that V is irreducible. Let $f, g \in k[x_1, \dots, x_n]$ be such that $f \cdot g \in I(V)$. Let $V_1 := V(f) \cap V$ and let $V_2 := V(g) \cap V$. By Lemma 1.37, we have that V_1 and V_2 are affine varieties. We first show that

$$V_1 \cup V_2 = V.$$

Clearly, $V_1 \cup V_2 \subset V$. Now assume that $x \in V$. Since $f \cdot g \in I(V)$, we have that $(f \cdot g)(x) = 0$. Thus, we have that $f(x) = 0$ or $g(x) = 0$, which implies that $x \in V_1$ or $x \in V_2$. Thus, our claim follows.

Since V is irreducible, it follows that $V_1 = V$ or $V_2 = V$. If $V_1 = V$ then $V \subset V(f)$, which implies that $f \in I(V)$. Similarly, if $V_2 = V$ then $g \in I(V)$. Thus, $I(V)$ is a prime ideal.

Assume now that $I(V)$ is a prime ideal and suppose by contradiction that $V = V_1 \cup V_2$ where $V_i \subset V$ is a proper affine variety. Since V_1 is proper, Lemma 1.38 implies that $V(I(V))$ is not contained in $V(I(V_1))$. Thus, there exists $f \in I(V_1)$ such that $f \notin I(V)$. Similarly, there exists $g \in I(V_2)$ such that $g \notin I(V)$. On the other hand, since $V = V_1 \cup V_2$, we have that $f \cdot g \in I(V)$, which contradicts the fact that $I(V)$ is a prime ideal. $\square$

1.4. Krull's lemma.

Proposition 1.43 (Krull's lemma). *Let R be a commutative ring and let $S \subset R$ be a multiplicative set (cf. Definition 1.4). Let $I \subset R$ be an ideal which is disjoint from S .*

Then there exists a prime ideal $\mathfrak{p} \subset R$ containing I and such that $\mathfrak{p}$ is disjoint from S .

Proof. Let P be the set of ideals in R containing I and disjoint from S . Then we can order the elements of P by inclusion. As in the proof of Proposition 1.28, we have that any totally ordered subset T of P admits an upper bound. Thus, by Zorn's Lemma (cf. Theorem 1.27), there exists a maximal element $\mathfrak{p}$ in P . In particular $\mathfrak{p}$ is disjoint from S . We need to show that $\mathfrak{p}$ is prime in R . Let $x, y \in R$ such that $x \cdot y \in \mathfrak{p}$. Assume by contradiction that $x, y \notin \mathfrak{p}$. Then, since $\mathfrak{p}$ is strictly included in $\mathfrak{p} + (x)$ and since $\mathfrak{p}$ is maximal in P , it follows that $\mathfrak{p} + (x)$ intersects S . Similarly, we have that $\mathfrak{p} + (y)$ intersects S and since S is multiplicative, we have that $(\mathfrak{p} + (x)) \cdot (\mathfrak{p} + (y))$ also intersects S . On the other hand, we have that

$$(\mathfrak{p} + (x)) \cdot (\mathfrak{p} + (y)) \subset \mathfrak{p} + (x \cdot y) = \mathfrak{p}.$$

Thus, $\mathfrak{p}$ intersects S , a contradiction. $\square$

Proposition 1.44. *Let R be a commutative ring and let $I \subset R$ be an ideal. Then*

$$\sqrt{I} = \bigcap_{I \subset \mathfrak{p}} \mathfrak{p}$$

where the intersection runs over all the prime ideals $\mathfrak{p}$ containing I .

Proof. If $\mathfrak{p}$ is a prime ideal containing I and $x \in R$ is such that $x^n \in I$ for some positive integer n , then $x^n \in \mathfrak{p}$ and, since $\mathfrak{p}$ is prime, it follows that $x \in \mathfrak{p}$. Thus, $\sqrt{I} \subset \mathfrak{p}$.

Assume now that $x \notin \sqrt{I}$. Then, we define

$$S := \{1, x, x^2, \dots\}$$

and it is easy to see that S is a multiplicative set which is disjoint from I . By Proposition 1.43, there exists a prime ideal $\mathfrak{p}$ of R containing I and which is disjoint from S . Thus, $x \notin \mathfrak{p}$. $\square$

Definition 1.45. Let R be a commutative ring. Then

- (1) the **nilradical** $\mathcal{N}(R)$ of R is the radical of the ideal (0) , i.e. $\mathcal{N}(R)$ is the set of all the nilpotent elements in R ;
- (2) the **Jacobson radical** $\mathcal{J}(R)$ of R is the intersection of all the maximal ideals of R .

It follows from Proposition 1.44 that the nilradical of a commutative ring R is the intersection of all the prime ideals of R .

1.5. Local rings.

Definition 1.46. Let R be a commutative ring. Then R is called a **local ring** if it has a unique maximal ideal $\mathfrak{m}$. The field $k := R/\mathfrak{m}$ is called the **residue field** of R (cf. Exercise 1.23).

A local ring R with a maximal ideal $\mathfrak{m}$ is often denoted as $(R, \mathfrak{m})$. In particular, if $(R, \mathfrak{m})$ is a local ring, we have that $\mathfrak{m} = \mathcal{J}(R)$.

Proposition 1.47. Let R be a local ring. Then the units in R are exactly the elements in $R \setminus \mathfrak{m}$.

Proof. The result follows easily from Proposition 1.28. $\square$

Proposition 1.48. Let R be a commutative ring and let $\mathfrak{m} \subset R$ be an ideal such that any $x \in R \setminus \mathfrak{m}$ is a unit. Then $(R, \mathfrak{m})$ is a local ring.

Proof. This is an easy exercise. $\square$

Example 1.49. If R is a field, then R is a local ring with maximal ideal $\mathfrak{m} = (0)$.

Let R be an integral domain, let $K = \text{Frac}(R)$ (cf. Proposition 1.5) and let $\mathfrak{p} \subset R$ be a prime ideal. Then Proposition 1.48 easily implies that

$$R_{\mathfrak{p}} := \left\{ \frac{f}{g} \in K \mid f, g \in R, g \notin \mathfrak{p} \right\}$$

is a local ring with maximal ideal

$$\mathfrak{m}_{\mathfrak{p}} := \left\{ \frac{f}{g} \in K \mid f, g \in R, f \in \mathfrak{p}, g \notin \mathfrak{p} \right\}.$$

The local ring $(R_{\mathfrak{p}}, \mathfrak{m}_{\mathfrak{p}})$ is called the **localisation** of R at $\mathfrak{p}$. Note that if $S := R \setminus \mathfrak{p}$ then S is multiplicative and $R_{\mathfrak{p}}$ coincides with $S^{-1}R$ (cf. Proposition 1.6).

Example 1.50. Let p be a prime in $\mathbb{Z}$. Then

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \in \mathbb{Q} \mid a, b \in \mathbb{Z}, p \nmid b \right\}.$$

Note that this is different from the ring

$$\mathbb{Z}[1/p] := \left\{ \frac{a}{p^k} \mid a \in \mathbb{Z}, k \geq 0 \right\}.$$

Example 1.51. Let k be a field. We then define the **formal power series ring** as

$$k[[x]] = \left\{ \sum_{i=0}^{\infty} a_i x^i \mid a_i \in k \right\}.$$

The elements of $k[[x]]$ are called *formal power series with coefficients in k* . Note that we do not impose any condition on the convergence of the series.

First, note that $R := k[[x]]$ is a ring, with the natural operations of sum and product. We now want to show that R is a local ring, with maximal ideal $\mathfrak{m} := (x)$. By Proposition 1.48, it is enough to show that any element in $R \setminus \mathfrak{m}$ is invertible. Let

$$f = \sum_{i=0}^{\infty} a_i x^i \in R \setminus \mathfrak{m} \quad a_0, a_1, \dots \in k.$$

Since $f \notin \mathfrak{m}$, it follows that $a_0 \neq 0$. Thus, we may write

$$f = a_0(1 + xg) \quad \text{where} \quad g = \sum_{i=0}^{\infty} a_{i+1} a_0^{-1} x^i \in R.$$

Then, if we define

$$f^{-1} := a_0^{-1}(1 - xg + x^2g^2 - \dots)$$

it is easy to check that $f^{-1} \in R$ is the inverse of f in R . Thus, our claim follows. Note that if f is a polynomial, then f^{-1} is the Taylor series of the function $1/f$ at 0.

1.6. Modules.

Definition 1.52. Let R be a commutative ring. A **R -module** (or a **module over R**) is an abelian group M with a multiplication map $R \cdot M \rightarrow M$ such that, for any $r, s \in R$ and for any $m, n \in M$, we have

- (1) $r \cdot (m \pm n) = r \cdot m \pm r \cdot n$;
- (2) $(r + s) \cdot m = r \cdot m + s \cdot m$;
- (3) $(rs) \cdot m = r \cdot (s \cdot m)$;
- (4) $1 \cdot m = m$.

Note that every abelian group is a $\mathbb{Z}$ -module and if R is a field then a R -module is a just a vector space over R .

Example 1.53. Let R be a ring and let $M = R[x]$ be the ring of polynomials in one variable. Then M is a R -module.

Definition 1.54. Let R be a commutative ring and M be a R -module. Then M is said to be **finitely generated** if there exist $m_1, \dots, m_n \in M$ such that for all $m \in M$ there exist $r_1, \dots, r_n \in R$ such that

$$m = \sum_{i=1}^n r_i m_i.$$

The elements $m_1, \dots, m_n$ are called **generators** of M over R .

If R is a commutative ring and $I \subset R$ is an ideal then I is a R -module. In particular, I is finitely generated if and only if there exist $m_1, \dots, m_n \in I$ such that $I = (m_1, \dots, m_n)$.

Definition 1.55. Let R be a commutative ring and let M be a R -module. A subset $N \subset M$ is called a **submodule** if $r \cdot m + s \cdot n \in N$ for all $m, n \in N$ and $r, s \in R$.

Note that if M is a module over a commutative ring R and $N \subset M$ is a submodule then the quotient M/N is also a module over R .

Example 1.56. Let R be a commutative ring and let M be a R -module. If $I \subset R$ is an ideal, then

$$I \cdot M := \left\{ \sum_{i=1}^k r_i \cdot m_i \mid r_1, \dots, r_k \in I, m_1, \dots, m_k \in M \right\}$$

is a submodule of M .

Note that differently from vector spaces, a submodule of a finitely generated module does not need to be finitely generated, as the following example shows:

Example 1.57. Let k be a field and let

$$R = k[x_1, x_2, \dots]$$

be the ring of polynomials in infinitely many variables over k . Then, $M := R$ is finitely generated as a module over itself by the element $1 \in R$. On the other hand, the ideal $I = (x_1, x_2, \dots) \subset R$ defines a submodule of M which is not finitely generated over R . Indeed, assume that $f_1, \dots, f_k \in I$. Then each f_i involves only finitely many variables and therefore there exists $q \geq 1$ such that $x_q \in I$ is not generated by $f_1, \dots, f_k$.

Definition 1.58. Let R be a commutative ring and let M and N be R -modules. Then a function

$$f: M \rightarrow N$$

is called **R -linear** (or a **morphism** of modules) if

$$f(r \cdot m + s \cdot n) = r \cdot f(m) + s \cdot f(n)$$

for all $m, n \in N$ and $r, s \in R$. The morphism f is called an **isomorphism** if it is a bijection

As in the case of rings, we will often refer to morphism of modules simply as morphisms.

Exercise 1.59. Let R be a commutative ring and let M be a R -module. We define the **annihilator** of M to be

$$\text{ann}(M) := \{r \in R \mid rm = 0 \text{ for all } m \in M\} \subset R.$$

- (1) Show that $\text{ann}(M) \subset R$ is an ideal.
- (2) Let $I \subset R$ be an ideal. Show that $I = \text{ann}(R/I)$.
- (3) Assume that M admits a unique generator $x \in M$ over R . Show that M is isomorphic to $R/\text{ann}(M)$.

Definition 1.60. Let R be a commutative ring and let M be a module over R . Then M is **simple** if the only submodules of M are $\{0\}$ and M .

Proposition 1.61. Let R be a commutative ring and let M be a module over R . Then M is simple if and only if M is isomorphic to $R/\mathfrak{m}$ for some maximal ideal $\mathfrak{m} \subset R$.

Proof. If $\mathfrak{m} \subset R$ is a maximal ideal then $R/\mathfrak{m}$ is a field (cf. Exercise 1.23) and hence it cannot have non-trivial submodules. Thus, $R/\mathfrak{m}$ is a simple module.

Assume now that M is simple. Let $v \in M$ be a non-zero element. Then

$$R \cdot v := \{r \cdot v \mid r \in R\} \subset M$$

is a non-zero submodule of M . Thus, $R \cdot v = M$, i.e. v is a generator of M . By Exercise 1.59 it follows that if $I := \text{ann}(M)$, then M is isomorphic to R/I . In particular, R/I is a simple module over R . By Proposition 1.28 there exists a maximal ideal $\mathfrak{m} \subset R$ containing I . Then $\mathfrak{m}/I$ is a submodule of R/I and since R/I is simple, we must have $I = \mathfrak{m}$. Thus, M is isomorphic to $R/\mathfrak{m}$. $\square$

2. NOETHERIAN AND ARTINIAN RINGS

2.1. Noetherian rings.

Definition 2.1. A commutative ring R is called **Noetherian** if all its ideals are finitely generated.

Definition 2.2. Let R be a commutative ring and let

$$I_1 \subset I_2 \subset I_3 \subset \dots$$

be an increasing chain of ideals of R . Then the chain satisfies the **ascending chain condition** (in short **ACC**) if there exists a positive integer k such that $I_\ell = I_k$ for all $\ell \geq k$.

Proposition 2.3. Let R be a commutative ring. Then the following are equivalent:

- (1) R is Noetherian;
- (2) every increasing chain of ideals satisfies ACC;
- (3) every non-empty set of ideals has a maximal element.

Proof.

(1) $\Rightarrow$ (2): Let

$$I_1 \subset I_2 \subset I_3 \subset \dots$$

be an increasing chain of ideals of R . Then, it is easy to check that

$$I := \bigcup_{i \geq 1} I_i$$

is an ideal of R . Thus, since R is Noetherian, I is finitely generated and there exist $m_1, \dots, m_k \in I$ such that $I = (m_1, \dots, m_k)$. Let $\ell \geq 1$ such that $m_1, \dots, m_k \in I_\ell$. Then $I_i = I_\ell$ for any $i \geq \ell$.

(2) $\Rightarrow$ (3): Let S be a non-empty set of ideals. Assume by contradiction that S does not admit a maximal element. Then there exists an increasing chain of ideals in S

$$I_1 \subset I_2 \subset I_3 \subset \dots$$

such that $I_i \neq I_{i+1}$ for all i , contradicting the ACC assumption.

(3) $\Rightarrow$ (1): Let $I \subset R$ be an ideal and assume by contradiction that I is not finitely generated. Let $m_1 \in I$. Then $I \neq (m_1)$ and we may pick $m_2 \in I \setminus (m_1)$. Proceeding this way, we find an increasing chain of ideals

$$I_1 \subset I_2 \subset I_3 \subset \dots$$

which does not admit a maximal element, contradiction our assumption. $\square$

Example 2.4. Let k be a field. Then

- (1) k is a Noetherian ring.
- (2) Let $R = k[x_1, x_2, \dots]$ be the ring of polynomials in infinitely many variables. Then R is not Noetherian. Indeed, if

$$I_i = (x_1, \dots, x_i)$$

then

$$I_1 \subset I_2 \subset I_3 \subset \dots$$

is an increasing chain of ideals which does not satisfy ACC (see Example 1.57 for a different proof).

Lemma 2.5. Let R be a Noetherian ring and let $I \subset R$ be an ideal. Then there exists a positive integer m such that I contains the m -th power of $\sqrt{I}$.

Proof. Since R is Noetherian, $\sqrt{I}$ is finitely generated. Let $x_1, \dots, x_k$ be generators. Then $x_i^{n_i} \in I$ for some positive integer n_i , for all $i = 1, \dots, k$. Let

$$m := \sum_{i=1}^k (n_i - 1) + 1.$$

Then $(\sqrt{I})^m$ is generated by the products $x_1^{r_1} \cdots x_k^{r_k}$ where $r_1, \dots, r_k$ are non-negative integers such that $\sum_{i=1}^k r_i = m$. Then, by the definition of m , we have that $r_i \geq n_i$ for some $i = 1, \dots, k$. Thus, $x_1^{r_1} \cdots x_k^{r_k} \in I$ and our claim follows. $\square$

Proposition 2.6. *Let R be a Noetherian ring. Then the nilradical $\mathcal{N}(R)$ of R is nilpotent, i.e. there exists a positive integer m such that $\mathcal{N}(R)^m = (0)$.*

Proof. Let $I = (0)$. Recall that $\mathcal{N}(R) = \sqrt{I}$. Thus, Lemma 2.5 immediately implies the claim. $\square$

2.2. Hilbert basis theorem. Let R be a commutative ring. For any polynomial

$$f(x) = \sum_{i=0}^n a_i x^i \in R[x],$$

of degree n (i.e. $a_n \neq 0$), we define the **leading term** of f as

$$LT(f) := a_n.$$

Theorem 2.7. *Let R be a Noetherian ring. Then $R[x]$ is also Noetherian.*

Proof. Let $J \subset R[x]$ be a non-zero ideal. For any $n \geq 0$, we define

$$I_n := \{r \in R \mid \exists f \in J \text{ such that } \deg f = n \text{ and } LT(f) = r\}.$$

Then

$$I_0 \subset I_1 \subset I_2 \subset \dots$$

is an increasing chain of ideals in R . Indeed, if $f \in J$ then $xf \in J$, $\deg(xf) = \deg f + 1$ and $LT(f) = LT(xf)$.

Since R is Noetherian, there exists $n \geq 0$ such that $I_i = I_n$ for any $i \geq n$. Moreover, since R is Noetherian, it also follows that each I_i is finitely generated. In particular, we may write

$$I_i = (a_{i,1}, \dots, a_{i,\ell_i}) \quad \text{where } a_{i,1}, \dots, a_{i,\ell_i} \in I_i$$

for any $i \leq n$. Thus, for each $i = 1, \dots, n$ and $j = 1, \dots, \ell_i$, there exists $f_{i,j} \in J$ such that $\deg f_{i,j} = i$ and $LT(f_{i,j}) = a_{i,j}$.

We want to show that J is generated by the elements $f_{i,j}$, for $i = 1, \dots, n$ and $j = 1, \dots, \ell_i$. Let $f \in J$ and let $m = \deg f$. Then $a := LT(f) \in I_m$. We proceed by induction on m . If $m = 0$ then $f \in I_0$ and we are done.

We now assume that $m \geq 1$. We distinguish two cases:

(1) If $m \geq n$ then $I_m = I_n$ and therefore

$$a = \sum_{i=1}^{\ell_n} r_i a_{n,i} \quad \text{for some } r_1, \dots, r_{\ell_n} \in R.$$

We define

$$g := f - \sum_{i=1}^{\ell_n} r_i x^{m-n} f_{n,i}.$$

Then $g \in J$ and $\deg g < \deg f$.

(2) If $m < n$ then

$$a = \sum_{i=1}^{\ell_m} r_i a_{m,i} \quad \text{for some } r_1, \dots, r_{\ell_m} \in R.$$

We define

$$g := f - \sum_{i=1}^{\ell_m} r_i f_{m,i}.$$

Also in this case, we have $g \in J$ and $\deg g < \deg f$.

In both cases, we may assume by induction that g is generated by the elements $f_{i,j}$, for $i = 1, \dots, n$ and $j = 1, \dots, \ell_i$. Thus, the same holds for f and we are done. $\square$

Example 2.8. Let k be a field. Then Theorem 2.7 easily implies by induction that $k[x_1, \dots, x_n]$ is a Noetherian ring for all $n \geq 1$.

In general, a subring of a Noetherian ring is not always Noetherian, as the following example shows:

Exercise 2.9. Let k be a field. Then, by Hilbert's Theorem, $k[x, y]$ is a Noetherian ring. Show that if

$$R := \left\{ \sum_{i,j \geq 0} a_{i,j} x^i y^j \mid a_{i,j} \in k \text{ such that } a_{0,i} = 0 \text{ if } i > 0 \right\} \subset k[x, y]$$

then R is not Noetherian. Hint: show that if

$$I_n := (x, xy, \dots, xy^n) \subset R$$

then

$$I_0 \subset I_1 \subset I_2 \subset \dots$$

is an increasing chain of ideals of R which does not satisfy ACC.

Proposition 2.10. Let k be a field and let n be a positive integer. Then the set of varieties in $\mathbb{A}_k^n$ satisfies the **descending chain condition** (or **DCC**). In other words, if

$$\mathbb{A}_k^n \supset V_1 \supset V_2 \supset \dots$$

is a decreasing chain of varieties, then there exists $k \geq 1$ such that $V_i = V_k$ for all $i \geq k$.

Proof. As in Example 2.8 we have that $k[x_1, \dots, x_n]$ is a Noetherian ring. We have that

$$I(V_1) \subset I(V_2) \subset \dots$$

is an increasing chain of ideals in $k[x_1, \dots, x_n]$ and, by Proposition 2.3 it satisfies ACC, i.e. there exists $k \geq 1$ such that $I(V_i) = I(V_k)$ for all $i \geq k$. Thus, our claim follows immediately by Lemma 1.38 $\square$

2.3. Cayley-Hamilton's theorem and Nakayama's lemma. Many results in linear algebra extend to modules. For example, let R be a commutative ring and let

$$M := \text{Mat}(n, R)$$

be the set of $(n \times n)$ -matrices with coefficients in R . Then M is a module over R .

We denote by $I_n \in M$ the identity matrix. Given $A \in M$, then the **determinant** $\det A$ of A is defined in the usual way, as an element of R . In particular, there exists a matrix B such that

$$A \cdot B = B \cdot A = \det A \cdot I_n.$$

Here, B is the **adjoint** of A and it is defined as the matrix with (i, j) -entry given by $(-1)^{i+j} \det A_{j,i}$ where $A_{j,i}$ is the matrix obtained from A by deleting the j -th row and the i -th column. In particular, A is invertible if and only if $\det A$ is a unit in R .

For any positive integers i, j , we denote by $\delta_{i,j}$ the (i, j) -**Kronecker symbol**, i.e.

$$\delta_{i,j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

Theorem 2.11 (Cayley-Hamilton's Theorem). *Let R be a commutative ring, let $I \subset R$ be an ideal and let M be a finitely generated R -module. Let $\varphi: M \rightarrow M$ a homomorphism such that*

$$\varphi(M) \subset I \cdot M.$$

Then there exist $a_0, \dots, a_{n-1} \in I$ such that

$$\varphi^n(m) + a_{n-1} \cdot \varphi^{n-1}(m) + \dots + a_1 \cdot \varphi(m) + a_0 \cdot m = 0$$

for all $m \in M$.

In the theorem, φ^i denotes the i -fold composition of φ with itself.

Proof. We can consider M as a $R[x]$ -module, by setting

$$x \cdot m := \varphi(m) \quad \text{for } m \in M.$$

We need to show that there exist $a_0, \dots, a_{n-1} \in I$ such that if

$$f(x) := x^n + a_{n-1}x^{n-1} + \dots + a_0$$

then $f(x) \cdot m = 0$ for all $m \in M$.

Let $m_1, \dots, m_n$ be generators of M over R . By assumption, we have $\varphi(m_i) \in I \cdot M$ for all $i = 1, \dots, n$. Thus, we may write

$$\varphi(m_i) = \sum_{j=1}^n r_{i,j} m_j \quad \text{for some } r_{i,j} \in I.$$

It follows that

$$\sum_{j=1}^n (x\delta_{i,j} - r_{i,j}) \cdot m_j = 0 \quad \text{for } i = 1, \dots, n.$$

Thus, if we consider the matrix

$$A := (x\delta_{i,j} - r_{i,j})_{i,j} \in \text{Mat}(n, R[x])$$

and its adjoint matrix B , then we obtain

$$0 = B \cdot A \cdot \begin{pmatrix} m_1 \\ \cdots \\ m_n \end{pmatrix} = \det A \cdot m_j \quad \text{for } j = 1, \dots, n.$$

Thus, $\det A \in R[x]$ is such that

$$\det A \cdot m = 0 \text{ for all } m \in M,$$

and our claim follows. $\square$

Theorem 2.12 (Nakayama's Lemma). *Let R be a commutative ring, let $I \subset R$ be an ideal and let M be a finitely generated R -module such that $I \cdot M = M$. Then there exists $a \in I$ such that*

$$a \cdot m = m \quad \text{for all } m \in M.$$

Proof. Let $\varphi := \text{Id}: M \rightarrow M$ be the identity homomorphism. Then, by Theorem 2.11, there exist $a_0, \dots, a_{n-1} \in I$ such that

$$(1 + a_{n-1} + \cdots + a_0) \cdot m = 0$$

for all $m \in M$. Define $a := -a_{n-1} - \cdots - a_0 \in I$. Then our claim follows. $\square$

Corollary 2.13. *Let R be a commutative ring and let M be a finitely generated R -module. Let $\varphi: M \rightarrow M$ be a surjective morphism. Then φ is an isomorphism.*

Proof. As in the proof of Theorem 2.11, we can consider M as a $R[x]$ -module, by setting

$$x \cdot m := \varphi(m) \quad \text{for } m \in M.$$

Since φ is surjective, we have that if $I := (x)$ then $I \cdot M = M$. By Nakayama's lemma (cf. theorem 2.12), there exists $f \in I$ such that $f \cdot m = m$ for all $m \in M$. Since $f \in (x)$, there exist $a_1, \dots, a_n \in R$ such that

$$f(x) = a_1x + a_2x^2 + \cdots + a_nx^n.$$

Thus, we have

$$a_1\varphi(m) + a_2\varphi^2(m) + \cdots + a_n\varphi^n(m) = m \quad \text{for all } m \in M.$$

Assume that $m \in \text{Ker}\varphi$. Then $\varphi(m) = 0$ which implies that $m = 0$. Thus, φ is injective, as claimed. $\square$

Corollary 2.14. *Let R be a commutative ring and let M be a finite generated R -module. Let $I \subset R$ be an ideal such that $I \subset \mathcal{J}(R)$ (cf. Definition 1.45) and $I \cdot M = M$. Then $M = 0$.*

Proof. By Nakayama's lemma (cf. theorem [2.12](#)), there exists $a \in I$ such that $a \cdot m = m$ for all $m \in M$. Let $x := 1 - a$. Since $a \in I$ and since $\mathcal{J}(R)$ is the intersection of all the maximal ideals of R , it follows that $x \notin \mathfrak{m}$ for all maximal ideals $\mathfrak{m} \subset R$. Thus, Proposition [1.28](#) implies that $(x) = R$, and by Exercise [1.15](#) it follows that x is a unit in R . Thus,

$$m = x^{-1} \cdot x \cdot m = x^{-1} \cdot (1 - a) \cdot m = 0 \quad \text{for all } m \in M,$$

and our claim follows. $\square$

2.4. Artin rings.

Definition 2.15. Let R be a commutative ring and let

$$I_1 \supset I_2 \supset I_3 \supset \dots$$

be a decreasing chain of ideals of R . Then the chain satisfies the **descending chain condition** (in short **DCC**) if there exists a positive integer k such that $I_\ell = I_k$ for all $\ell \geq k$.

Definition 2.16. Let R be a commutative ring. Then R is called an **Artin ring** (or **Artinian**) if any decreasing chain of ideals of R satisfies the DCC.

Proposition 2.17. Let R be a commutative Artin ring. Then every prime ideal of R is maximal.

Proof. Let $\mathfrak{p} \subset R$ be a prime ideal. Then, as in Exercise [1.30](#) we have that $S := R/\mathfrak{p}$ is an integral domain. It is easy to see that it is an Artin ring.

By Exercise [1.23](#) it is enough to show that S is a field. Let $0 \neq x \in S$. Then there exists a positive integer n such that $(x^n) = (x^{n+1})$ and hence $x^n = x^{n+1} \cdot y$ for some $y \in S$. Since S is an integral domain and $x \neq 0$, it follows that $xy = 1$. Thus, x has an inverse in S , as claimed. $\square$

Recall from Definition [1.45](#) that if R is a commutative ring, then the nilradical $\mathcal{N}(R)$ of R is the radical of the ideal (0) , i.e. $\mathcal{N}(R)$ is the set of all the nilpotent elements in R ; the Jacobson radical $\mathcal{J}(R)$ of R is the intersection of all the maximal ideals of R . We have

Corollary 2.18. Let R be a commutative Artin ring. Then

$$\mathcal{N}(R) = \mathcal{J}(R).$$

Proof. By Proposition [1.44](#), it follows that $\mathcal{N}(R)$ is the intersection of all the prime ideals of R . Thus, Proposition [2.17](#) implies the claim. $\square$

Let us assume now that R is an Artin local ring with maximal ideal $\mathfrak{m}$. Then, by Proposition [2.17](#) it follows that $\mathfrak{m}$ is the only prime ideal of R and Corollary [2.18](#) implies that every element of $\mathfrak{m}$ is nilpotent. Thus, by Proposition [1.47](#) we have that the elements of R are either nilpotent or they are units.

Example 2.19.

- (1) As in Example 1.49, every field is an Artin local ring.
- (2) If $p \in \mathbb{Z}$ is a prime number and n is a positive integer, then $\mathbb{Z}/(p^n)$ is a local Artin ring.
- (3) Similarly if k is a field and n is a positive integer then $k[x]/(x^n)$ is a local Artin ring.

2.5. Dimension of a ring. In §2.1 and §2.4 we have defined Noetherian and Artin rings. We now extend these definitions to modules:

Definition 2.20. Let R be a commutative ring and let M be a module over R . Then

- (1) M is **Noetherian** if any increasing sequence of submodules

$$M_1 \subset M_2 \subset \cdots \subset M$$

satisfies ACC.

- (2) M is **Artinian** if any decreasing sequence of submodules

$$M \supset M_1 \supset M_2 \supset \cdots$$

satisfies DCC.

Example 2.21. Let k be a field. Recall that a k -module is just a vector space over k . Thus, the following are equivalent

- (1) M is a finite dimensional vector space;
- (2) M is Noetherian;
- (3) M is Artinian.

Exercise 2.22. Let R be a commutative ring and let $I \subset R$ be an ideal. Show that

- (1) R is a Noetherian ring if and only if I and R/I are Noetherian R -modules.
- (2) R is an Artin ring if and only if I and R/I are Artinian R -modules.

Exercise 2.23. Let R be a ring and let M be a R -module. Let $P \subset N \subset M$ be submodules such that the natural map

$$N/P \rightarrow M/P$$

is an isomorphism. Show that $N = M$.

Definition 2.24. (1) Let X be a topological space. The **dimension** of X is

$$\dim X := \sup \{n \geq 0 : \exists X_0 \subsetneq X_1 \subsetneq X_2 \cdots \subsetneq X_n \subset X \text{ with } X_i \text{ closed and irreducible} \}.$$

- (2) Let R be a commutative ring. The **dimension** of R is

$$\dim R := \sup \{n \geq 0 : \exists \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \mathfrak{p}_2 \cdots \subsetneq \mathfrak{p}_n \subset R \text{ with } \mathfrak{p}_0, \dots, \mathfrak{p}_n \text{ prime ideals} \}$$

- (3) Let R be a commutative ring and let $\mathfrak{p} \subset R$ be a prime ideal. The **height** of $\mathfrak{p}$ is

$$h(\mathfrak{p}) := \sup \{n \geq 0 : \exists \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \mathfrak{p}_2 \cdots \subsetneq \mathfrak{p}_n = \mathfrak{p} \text{ with } \mathfrak{p}_0, \dots, \mathfrak{p}_n \text{ prime ideals} \}$$

Example 2.25.

- (1) $\dim \mathbb{Z} = 1$ (cf. Example 1.32).

- (2) Let k be a field. Then $\dim k = 0$.
 (3) Let k be a field and let $R = k[x_1, \dots, x_n]$. Then we have a chain of prime ideals

$$(0) \subsetneq (x_1) \subsetneq (x_1, x_2) \subsetneq \cdots \subsetneq (x_1, \dots, x_n) \subsetneq R.$$

Thus, $\dim R \geq n$. We will show later that $\dim R = n$ (cf. Corollary 4.10).

Definition 2.26. Let R be a commutative ring and let M be a module over R . Then a chain of submodules

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

is called a **Jordan-Hölder series** (or **composition series**) if each module M_i/M_{i-1} is a simple R -module for $i = 1, \dots, n$ (cf. Definition 1.60). The number n is called the **length** of the Jordan-Hölder series.

Proposition 2.27. Let R be a commutative ring and let M be a module over R . Let

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

be a Jordan-Hölder series. Then every Jordan-Hölder series has length n .

Proof. Given a module M which admits a Jordan-Hölder series, we denote by $\ell(M)$ the minimum length of a Jordan-Hölder series in M .

We first show that if $N \subset M$ is a submodule, then N admits a Jordan-Hölder series and $\ell(N) \leq \ell(M)$. Let

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

be a Jordan-Hölder series in M of minimum length and let $N_i := N \cap M_i$ for $i = 0, \dots, n$. Then

$$N_i/N_{i-1} \subset M_i/M_{i-1} \quad \text{for } i = 1, \dots, n.$$

Since M_i/M_{i-1} is simple, it follows that either $N_i = N_{i-1}$ or $N_i/N_{i-1} = M_i/M_{i-1}$ is simple. Hence, removing the repeated terms, we get a Jordan-Hölder series of length at most n . Note that if the equality holds, then Exercise 2.23 implies that $N_i = M_i$ for all $i = 0, \dots, n$ and, in particular, $N = M$.

Let

$$\{0\} = M'_0 \subsetneq M'_1 \subsetneq \cdots \subsetneq M'_k = M$$

be a Jordan-Hölder series of M . Then, by what we have proven above, it follows that

$$0 = \ell(M'_0) < \ell(M'_1) < \cdots < \ell(M'_k) = \ell(M).$$

Thus, $k \leq \ell(M)$. Since $\ell(M)$ is the minimum length of a Jordan-Hölder series in M , the equality holds. Thus, our claim follows. $\square$

Definition 2.28. Let R be a commutative ring and let M be a module over R . If M admits a Jordan-Hölder series then we define the **length** $\ell(M)$ of M to be the length of any Jordan-Hölder series of M (note that by Proposition 2.27, this number is well-defined). If M does not admit any Jordan-Hölder series then we write $\ell(M) = +\infty$.

Proposition 2.29. *Let R be a commutative ring and let M be a module over R such that $\ell(M) < +\infty$. Then any chain of submodules*

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

can be extended to a Jordan-Hölder series.

Proof. If $n \geq \ell(M)$, then as in the proof of Proposition 2.27 it follows that $n = \ell(M)$ and that

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

is a Jordan-Hölder series.

If $n < \ell(M)$, then

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

is not a Jordan-Hölder series and therefore new submodules can be inserted until the length is $\ell(M)$. $\square$

Proposition 2.30. *Let R be a commutative ring and let M be a module over R . Then $\ell(M) < +\infty$ if and only if M is Noetherian and Artinian.*

Proof. Assume that $\ell(M) < +\infty$ then any chains of submodules in M is of bounded length and, therefore, both ACC and DCC hold. Thus M is Noetherian and Artinian.

Assume now that M is Noetherian and Artinian. We aim to construct a Jordan-Hölder series. Since M is Noetherian, by Zorn's Lemma (cf. Theorem 1.27), there exists a maximal submodule $M^1 \subsetneq M$. In particular, it follows that M/M^1 is simple. If $M^1 = \{0\}$ then we are done. Otherwise, using Zorn's Lemma again, there exists $M^2 \subsetneq M^1$ maximal submodule such that M^1/M^2 is simple. By continuing this way, we have a decreasing sequence

$$M \supsetneq M^1 \supsetneq M^2 \supsetneq \dots$$

By the Artinian property, the sequence satisfies DCC. Thus, it is a Jordan-Hölder series and our claim follows. $\square$

2.6. Primary decomposition.

Definition 2.31. *Let R be a commutative ring and let $I \subsetneq R$ be a proper ideal. Then*

- (1) *I is said to be **irreducible** if for any ideals $J_1, J_2 \subset R$ such that $I = J_1 \cap J_2$, we have that $I = J_1$ or $I = J_2$.*
- (2) *I is said to be **primary** if, for any $x, y \in R$ such that $x \cdot y \in I$, we have that $x \in I$ or $y^n \in I$ for some $n > 0$.*

Proposition 2.32. *Let R be a commutative ring and let $I \subset R$ be a proper ideal*

- (1) *If I is prime then I is primary.*
- (2) *I is primary if and only if every zero divisor in R/I is nilpotent.*

Proof. This is an easy exercise. $\square$

Proposition 2.33. *Let R be a commutative ring and let $I \subset R$ be a primary ideal. Then $\sqrt{I}$ is the smallest prime ideal containing I .*

Proof. By Proposition 1.44, it is enough to show that $\sqrt{I}$ is a prime ideal. Let $x, y \in R$ such that $x \cdot y \in \sqrt{I}$. Then $(x \cdot y)^m \in I$ for some positive integer m . Thus, $x^m \in I$ or $y^{mn} \in I$ for some positive integer n , which implies that $x \in \sqrt{I}$ or $y \in \sqrt{I}$. $\square$

Definition 2.34. *Let R be a commutative ring and let $I \subset R$ be a primary ideal. Then, if $\mathfrak{p} := \sqrt{I}$, we say that I is **$\mathfrak{p}$ -primary**.*

Exercise 2.35.

- (1) Show that if $R = \mathbb{Z}$ then the primary ideals are (0) and (p^n) where $p \in \mathbb{Z}$ is a prime number and n is a positive integer.
- (2) Let k be a field, let $R = k[x, y]$ and let $I = (x, y^2)$. Show that I is primary.

Proposition 2.36. *Let R be a commutative ring and let $I \subset R$ be an ideal such that $\sqrt{I}$ is maximal. Then I is primary.*

Proof. Let $\mathfrak{m} = \sqrt{I}$. Then Proposition 1.44 implies that $\mathfrak{m}$ is the only prime ideal containing I . Thus, R/I has exactly one prime ideal and Exercise 1.35 implies that every element of R/I is either a unit or nilpotent. Thus, every zero divisor of R/I is nilpotent and Proposition 2.32 implies that I is primary. $\square$

Definition 2.37. *Let R be a commutative ring and let $I \subset R$ be an ideal. A **primary decomposition** of I is an expression of I as a finite intersection of primary ideals, i.e.*

$$I = \bigcap_{i=1}^n J_i$$

with $J_1, \dots, J_n \subset R$ primary ideals.

Lemma 2.38. *Let R be Noetherian ring and let $I \subset R$ be an ideal. Then I is a finite intersection of irreducible ideals.*

Proof. Consider the set

$$P := \{J \subset R \text{ ideal} \mid J \text{ is not a finite intersection of irreducible ideals}\}.$$

Assume by contradiction that P is not empty. Then since R is Noetherian, Zorn's lemma (cf. Theorem 1.27) implies that P admits a maximal element J_0 with respect to the inclusion. In particular, J_0 is not irreducible and there exist ideals J_1 and J_2 such that

$$J_0 = J_1 \cap J_2 \quad \text{and} \quad J_0 \neq J_i \quad i = 1, 2.$$

Thus, by maximality, $J_i \notin P$ and therefore it is a finite intersection of irreducible ideals, for $i = 1, 2$. But then also J_0 is a finite intersection of irreducible ideals, a contradiction. $\square$

Lemma 2.39. *Let R be a Noetherian ring and let $I \subset R$ be an irreducible ideal. Then I is primary.*

Proof. Let $S := R/I$. Since I is irreducible, it follows easily that $(0) \subset S$ is an irreducible ideal. By Proposition 2.32, it is enough to show that every zero divisor x of S is nilpotent. Let $y \neq 0$ such that $x \cdot y = 0$ and let $J_i := \text{ann}(x^i)$ (cf. Exercise 1.59) for all $i \geq 1$. Then

$$J_1 \subset J_2 \subset \dots$$

is an ascending chain of ideals and by ACC, there exists a positive integer n such that $J_i = J_n$ for all $i \geq n$.

We want to show that

$$(1) \quad (x^n) \cap (y) = (0).$$

Indeed, let $r \in (x^n) \cap (y)$. Since $r \in (y)$ we have that $r \cdot x = 0$ and since $r \in (x^n)$ we have that $r = s \cdot x^n$ for some $s \in S$ and hence $s \cdot x^{n+1} = r \cdot x = 0$. Thus, $s \in J_{n+1} = J_n$, which implies that $r = s \cdot x^n = 0$ and (1) follows.

Since $(0) \subset S$ is irreducible and $(y) \neq (0)$, it follows that $(x^n) = (0)$, i.e. $x^n = 0$. Therefore, x is nilpotent and our claim follows. $\square$

Theorem 2.40. *Let R be a Noetherian ring and let $I \subset R$ be an ideal. Then I admits a primary decomposition.*

Proof. The Theorem follows immediately from Lemma 2.38 and Lemma 2.39. $\square$

Lemma 2.41. *Let R be a commutative ring and assume that*

$$(0) = \mathfrak{m}_1 \cdot \dots \cdot \mathfrak{m}_n$$

where $\mathfrak{m}_1, \dots, \mathfrak{m}_n \subset R$ are maximal ideals. Then R is Noetherian if and only if R is Artinian.

Proof. Let $k_i := R/\mathfrak{m}_i$. By Exercise 1.23, we have that k_i is a field. Consider the chain

$$(0) = \mathfrak{m}_1 \cdot \dots \cdot \mathfrak{m}_n \subset \dots \subset \mathfrak{m}_1 \cdot \mathfrak{m}_2 \subset \mathfrak{m}_1 \subset R.$$

Then, each quotient $M_i := (\mathfrak{m}_1 \cdot \dots \cdot \mathfrak{m}_{i-1})/(\mathfrak{m}_1 \cdot \dots \cdot \mathfrak{m}_i)$ is a module over k_i , and therefore a vector space, for each $i = 1, \dots, n$. Thus, as in Example 2.21 we have that M_i is Noetherian if and only if it is Artinian. The claim then follows by repeated applications of Exercise 2.22. $\square$

Theorem 2.42. *Let R be a Noetherian ring. Then $\dim R = 0$ if and only if R is an Artin ring.*

Proof. Assume that R is Artinian. Then Proposition 2.17 implies that every prime ideal is maximal. Thus, $\dim R = 0$.

Assume now that $\dim R = 0$. By Theorem 2.40, the ideal (0) admits a primary decomposition, i.e.

$$(0) = \bigcap_{i=1}^n J_i$$

with $J_1, \dots, J_n \subset R$ primary ideals.

Let $\mathfrak{p} \subset R$ be a prime ideal. We want to show that there exists $i = 1, \dots, n$ such that $J_i \subset \mathfrak{p}$. Suppose not. Then, for all $i = 1, \dots, n$, there exists $x_i \in J_i \setminus \mathfrak{p}$. We have

$$x_1 \cdots x_n \in \bigcap_{i=1}^n J_i = (0).$$

Thus, $x_1 \cdots x_n \in (0) \subset \mathfrak{p}$. Since $\mathfrak{p}$ is prime, we have that $x_i \in \mathfrak{p}$ for some $i = 1, \dots, n$, a contradiction. Thus, by Zorn's lemma (cf. Theorem 1.27), it follows that R admits a minimal prime ideal $\mathfrak{p}$, i.e. a prime ideal $\mathfrak{p}$ such that if $J \subset R$ is a prime ideal contained in $\mathfrak{p}$ then $J = \mathfrak{p}$. Moreover, any minimal prime ideal coincides with $\sqrt{J_i}$ for some $i = 1, \dots, n$. Thus, R admits finitely many minimal prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_k$. Since $\dim R = 0$, it follows that $\mathfrak{p}_1, \dots, \mathfrak{p}_k$ are also maximal ideals. By Proposition 2.6, we have that the nilradical $\mathcal{N}(R)$ of R is nilpotent, i.e. there exists a positive integer q such that $\mathcal{N}(R)^q = 0$. On the other hand, Proposition 1.44 implies that

$$\mathcal{N}(R) = \bigcap_{i=1}^k \mathfrak{p}_i.$$

Thus, we have

$$\mathfrak{p}_1^q \cdots \mathfrak{p}_k^q \subset \left(\bigcap_{i=1}^k \mathfrak{p}_i \right)^q = \mathcal{N}(R)^q = (0).$$

By Lemma 2.41, since R is Noetherian, it follows that R is Artinian. $\square$

Proposition 2.43. *Let $(R, \mathfrak{m})$ be a Noetherian local ring. Then exactly one of the following is true:*

- (1) $\mathfrak{m}^n \neq \mathfrak{m}^{n+1}$ for all $n \geq 0$; or
- (2) $\mathfrak{m}^n = 0$ for some n and A is an Artin ring.

Proof. Assume that $\mathfrak{m}^n = \mathfrak{m}^{n+1}$ for some $n \geq 0$. Since $\mathfrak{m} = \mathcal{J}(R)$, Corollary 2.14 implies that $\mathfrak{m}^n = 0$. Let $\mathfrak{p}$ be any prime ideal of R . Since $\mathfrak{m}^n = 0 \subset \mathfrak{p}$, we must have $\mathfrak{m} = \mathfrak{p}$, i.e. $\mathfrak{m}$ is the only prime ideal of R . Thus, $\dim R = 0$ and Theorem 2.42 implies that R is an Artin ring. $\square$

Exercise 2.44. *Let R be a commutative ring. Show that R is Artinian if and only if R is Noetherian and $\dim R = 0$.*

3. NULLSTELLENSATZ

3.1. Algebras.

Definition 3.1. *Let R be a commutative ring. A R -algebra (or algebra over R) is a ring A which admits a morphism of rings $\varphi: R \rightarrow A$.*

Note that every R -algebra is a R -module, where the operation is defined by $r \cdot a = \varphi(r)a$, for any $r \in R$ and $a \in A$.

Example 3.2. If S is a commutative ring and $R \subset S$ is a subring then S is an R -algebra, which is called an **extension** of R .

Definition 3.3. Let R be a commutative ring and let $\varphi: R \rightarrow A$ be a R -algebra. Then

- (1) A is said to be **finite** over R if it is finitely generated as an R -module.
- (2) an element $a \in A$ is said to be **integral** over A if there exists a monic polynomial

$$f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in R[x]$$

such that $f(a) = 0$.

- (3) A is said to be **integral** over R if every $a \in A$ is integral.
- (4) Let $S \subset A$ be a subset. Then we denote by $R[S] \subset A$ the **R -algebra generated** by S , i.e. the smallest subring of A containing $\varphi(R)$ and S .

Note that there is a big difference between being generated as an algebra or as a module, as shown in the following example:

Example 3.4. Let k be a field and let $A := k[x]$. Then A is a k -algebra and, therefore, it is also a k -module. The k -module generated by x is the k -module

$$\{a \cdot x \mid a \in k\} \subset k[x].$$

On the other hand, the k -algebra generated by x is the whole algebra A .

Lemma 3.5. Let R, S and T be commutative rings such that $R \subset S \subset T$. Assume that T is a finite S -algebra and S is a finite R -algebra.

Then T is a finite R -algebra.

Proof. This is an easy exercise. □

Proposition 3.6. Let R be a commutative ring, let $\varphi: R \rightarrow A$ be an algebra over R and let $a \in A$. Then a is integral over R if and only if $R[a]$ is finite over R .

Proof. Assume that a is integral over R . Then $R[a]$ is the R -module generated by $1, a, \dots, a^{n-1}$. Thus, $R[a]$ is finite over R .

Assume now that $R[a]$ is finite over R . Then there exist $r_{i,j} \in R$ with $i = 1, \dots, m$ and $j = 1, \dots, n_i$ such that if

$$f_i = \sum_{j=0}^{n_i} \varphi(r_{i,j})a^j \in R[a] \quad \text{for } i = 1, \dots, m$$

then $f_1, \dots, f_m$ generate $R[a]$. Let $n = \max\{n_1, \dots, n_m\}$. Then, since $a^{n+1} \in R[a]$, it follows that

$$a^{n+1} = \sum_{i=0}^n \varphi(s_i)a^i$$

for some $s_0, \dots, s_n \in R$. Thus, a is integral over R . □

Proposition 3.7. *Let $R \subset S$ be commutative rings. Then S is finite over R if and only if $S = R[a_1, \dots, a_n]$ for some $a_1, \dots, a_n \in S$ which are integral over R .*

Moreover, in this case, S is integral over R .

Proof. Let $a_1, \dots, a_n$ be integral over R . Then by Proposition 3.6 and Lemma 3.5, it follows immediately that $S = R[a_1, \dots, a_n]$ is finite over R .

Assume now that S is finite over R and let $a_1, \dots, a_n$ be generators of S over R . We will show that every element of S is integral over R , so that, S is integral over R and, in particular, $a_1, \dots, a_n$ are integral over R .

Let $a \in S$ and consider the R -linear morphism

$$\varphi: S \rightarrow S \quad s \mapsto a \cdot s.$$

Then by Cayley-Hamilton theorem (cf. Theorem 2.11) there exist $c_0, \dots, c_{k-1} \in R$ such that

$$\varphi^k(s) + c_{k-1}\varphi^{k-1}(s) + \dots + c_0 \cdot s = 0$$

for all $s \in S$. In particular, if $s = 1$ then

$$a^k + c_{k-1}a^{k-1} + \dots + a_0 = 0.$$

Thus, a is integral over R , as claimed. $\square$

Example 3.8.

- (1) Let k be a field and let

$$k[x^2] := \left\{ \sum_{i=0}^n a_i x^{2i} \mid a_i \in k \right\}.$$

Then $k[x]$ is an extension of $k[x^2]$, which is integral and finite.

- (2) Let k be a field, let $R = k[x]$ and consider the R -algebra

$$A = k[x, x^{1/2}, x^{1/4}, x^{1/8}, \dots]$$

which is the ring obtained by adjoining all 2^n -th roots of x to R . Then A is integral over R but it is not finite over R .

- (3) Let R be a commutative ring of characteristic p , i.e. p is a prime number such that $p \cdot 1 = 0$. Then the **Frobenius morphism** is defined as

$$F: R \rightarrow R \quad r \mapsto r^p.$$

Recall that R is said to be **perfect** if F is surjective.

Assume now that k is a field of characteristic p and $R = k[x]$ then R is an integral extension of the image $F(R)$.

- (4) Recall from Example 1.50 that if p is a prime, we defined

$$\mathbb{Z}[1/p] := \left\{ \frac{a}{p^k} \mid a \in \mathbb{Z}, k \geq 0 \right\}.$$

Then $\mathbb{Z}[1/p]$ is a $\mathbb{Z}$ -algebra which is not integral.

3.2. Normal rings and integral closure. The following result is a variation of Lemma 3.5:

Lemma 3.9. *Let R, S and T be commutative rings such that $R \subset S \subset T$. Assume that T is integral over S and S is integral over R .*

Then T is integral over R .

Proof. Let $t \in T$. Since T is integral over S , there exist $b_0, \dots, b_{n-1} \in S$ such that

$$t^n + b_{n-1}t^{n-1} + \dots + b_0 = 0.$$

Thus, t is integral over $R[b_0, \dots, b_{n-1}]$.

Since S is integral over R , we have that $b_0, \dots, b_{n-1}$ are integral over R . Thus, Proposition 3.6 implies that $R[b_0, \dots, b_{n-1}, t]$ is finite over $R[b_0, \dots, b_{n-1}]$ and that $R[b_0, \dots, b_{n-1}]$ is finite over R . By Lemma 3.5, it follows that $R[b_0, \dots, b_{n-1}, t]$ is finite over R . Thus, applying Proposition 3.6 again, we have that t is integral over R . $\square$

Proposition 3.10. *Let R be a commutative ring and let A be an algebra over R . Then the subset*

$$\tilde{R} := \{a \in A \mid a \text{ is integral over } R\} \subset A$$

is a subring of A . Moreover, if $b \in A$ is integral over $\tilde{R}$ then $b \in \tilde{R}$.

Proof. Let $a, b \in \tilde{A}$ then $a \pm b, a \cdot b \in R[a, b]$. By Proposition 3.7 we have that $R[a, b]$ is finite over R . Thus, Proposition 3.6 implies that $a \pm b, a \cdot b \in R$ and, therefore, $\tilde{R}$ is a subring of A .

The second part of the proposition follows immediately from Lemma 3.9. $\square$

Definition 3.11. *Let R be a commutative ring and let A be an algebra over R . The ring $\tilde{R}$ in Proposition 3.10 is called the **integral closure** of R in A . If $\tilde{R} = R$ then R is called **integrally closed** in A .*

*Assume now that R is an integral domain and let $A = \text{Frac}(R)$ (cf. Proposition 1.5). The integral closure of R in A is called the **normalisation** of R . If R is integrally closed in A then R is called **normal**.*

Exercise 3.12. *Show that any UFD (cf. 1.18) is normal.*

Example 3.13. *Recall that a number field $K \supset \mathbb{Q}$ is a finite extension of the rational field $\mathbb{Q}$, i.e. an extension of $\mathbb{Q}$ which is a finite dimensional vector space over $\mathbb{Q}$. In this case, the integral closure of $\mathbb{Z}$ in K is called the **ring of integers** $\mathcal{O}_K$ of K .*

For example, it follows from elementary number theory that if n is a square-free positive integer (i.e. a product of distinct prime numbers) and $K = \mathbb{Q}(\sqrt{n})$ is the field extension generated by $\sqrt{n}$ then the ring of integers of K is

$$\mathcal{O}_K = \mathbb{Z}[\alpha]$$

where

$$\alpha = \begin{cases} \frac{1+\sqrt{n}}{2} & \text{if } n \equiv 1 \pmod{4} \\ \sqrt{n} & \text{if } n \equiv 2 \text{ or } 3 \pmod{4}. \end{cases}$$

Example 3.14. Let k be a field and let

$$R := k[x, y]/(y^2 - x^3).$$

Then the field of fraction of R is

$$\text{Frac}(R) = k(t) \quad \text{where } t = \frac{y}{x}.$$

We have

$$t^2 - x = \frac{y^2 - x^3}{x^2} = 0 \quad \text{in } R.$$

On the other hand, t is not contained in R . Thus R is not normal.

From Exercise 3.12, it follows that $k[t]$ is normal. Thus, $k[t]$ is the normalisation of R .

3.3. Noether normalisation lemma. Recall that if k is a field then a polynomial $f(x_1, \dots, x_n) \in k[x_1, \dots, x_n]$ is called **homogeneous** if it is a sum of monomial of the same degree. We begin with the following simple remark:

Remark 3.15. Let k be an infinite field and let $f(x_1, \dots, x_n) \in k[x_1, \dots, x_n]$ be a non-zero homogeneous polynomial. Then we claim that there exist $c_1, \dots, c_{n-1} \in k$ such that

$$f(c_1, \dots, c_{n-1}, 1) \neq 0.$$

Indeed, we proceed by induction on n . If $n = 1$ then f is a non-zero multiple of a monomial and therefore $f(1) \neq 0$. Assume now that $n > 1$. We may write

$$f(x_1, \dots, x_n) = \sum_{i=0}^d x_1^i f_i(x_2, \dots, x_n)$$

for some homogeneous polynomial $f_i \in k[x_2, \dots, x_n]$ of degree $d - i$. Since f is non-zero, we may assume that f_i is non-zero for some $i = 0, \dots, d$. By induction, there exist $c_2, \dots, c_{n-1}$ such that

$$f_i(c_2, \dots, c_{n-1}, 1) \neq 0.$$

Let

$$g(x) := f(x, c_2, \dots, c_{n-1}, 1) \in k[x].$$

Then g is a non-zero polynomial in one variable and, in particular, it has only finitely many zeroes. Since k is infinite, there exist $c_1 \in k$ such that

$$f(c_1, c_2, \dots, c_{n-1}, 1) \neq 0,$$

as claimed.

Note that the result is not true over a finite field. For example, consider $k := \mathbb{Z}/p\mathbb{Z}$ for some prime p and let

$$f(x_1, x_2) := x_1^p - x_2^p \in k[x_1, x_2].$$

Then, by Fermat little theorem, $f(c, 1) = 0$ for all $c \in k$.

Definition 3.16. Let k be a field and let A be an algebra over k . Then the elements $a_1, \dots, a_n \in A$ are said to be **algebraically independent over k** if there does not exist a non-zero polynomial $F \in k[x_1, \dots, x_n]$ such that

$$F(a_1, \dots, a_n) = 0.$$

Equivalently, the morphism

$$k[x_1, \dots, x_n] \rightarrow k[a_1, \dots, a_n] \quad x_i \mapsto a_i \quad \text{for } i = 1, \dots, n$$

is an isomorphism. ^[4]

The following theorem states that if A is a finitely generated algebra over a field k then we can find a decomposition

$$k \subset k[a_1, \dots, a_n] \subset A$$

where $k[a_1, \dots, a_n]$ is a polynomial extension of k and $k[a_1, \dots, a_n] \subset A$ is a finite extension.

Theorem 3.17 (Noether normalisation^[5] Lemma). Let k be a field and let A be a finitely generated k -algebra. Then there exist $a_1, \dots, a_n \in A$ such that

- (1) $a_1, \dots, a_n$ are algebraically independent over k ;
- (2) A is finite over $k[a_1, \dots, a_n]$.

Proof. We first assume that k is an infinite field, as in this case the proof is easier.

Let $z_1, \dots, z_m$ be generators of A over k . We proceed by induction on m . Assume first that $m = 1$. If z_1 is integral over k then Proposition 3.7 implies that A is finite over k and, thus, we are done. If z_1 is not integral over k then z_1 is algebraically independent and we can choose $a_1 = z_1$ so that $A = k[a_1]$.

We now assume that $m > 1$. If $z_1, \dots, z_m$ are algebraically independent over k then we may choose $n = m$ and $a_i = z_i$, for $i = 1, \dots, n$ and we are done. Thus, we may assume that there exists a non-zero polynomial $f \in k[x_1, \dots, x_m]$ such that

$$f(z_1, \dots, z_m) = 0.$$

Let $d = \deg f$ and let f_d be the homogenous part of f of degree d . Since k is infinite, by Remark 3.15 we may find $c_1, \dots, c_{m-1}$ such that

$$f_d(c_1, \dots, c_{m-1}, 1) \neq 0.$$

⁴Note that the definition might be slightly confusing. Indeed, if k is a field and A is a k -algebra, then an element $a \in A$ is algebraically independent if and only if it is not integral over k .

⁵Note that the notion of 'normalisation' here is different from the one in Definition 3.11.

Define

$$y_i := z_i - c_i z_m \quad \text{for } i = 1, \dots, m-1.$$

Then

$$\begin{aligned} 0 &= f(z_1, \dots, z_m) = f(y_1 + c_1 z_m, \dots, y_{m-1} + c_{m-1} z_m, z_m) \\ &= f_d(c_1, \dots, c_{m-1}, 1) z_m^d + g_1 z_m^{d-1} + \dots + g_d \end{aligned}$$

where $g_i \in k[y_1, \dots, y_{m-1}]$ is a polynomial of degree i . Thus, it follows that z_m is integral over $k[y_1, \dots, y_{m-1}]$. By Proposition 3.10, we have that $z_i = y_i + c_i z_m$ is also integral over $k[y_1, \dots, y_{m-1}]$, for $i = 1, \dots, m-1$. Thus, Proposition 3.7 implies that A is finite over $k[y_1, \dots, y_{m-1}]$. The Theorem follows by using induction and Lemma 3.5.

We now consider the general case (with k not necessarily finite). As above, we let $z_1, \dots, z_m$ be generators of A over k and we proceed by induction on m and, as above, we may assume that $m > 1$ and that there exists a non-zero polynomial $f \in k[x_1, \dots, x_m]$ such that

$$f(z_1, \dots, z_m) = 0.$$

Pick some positive integers $r_1, \dots, r_m$ (we will impose some extra assumptions on the r_i 's later on) and let

$$y_i = x_i - x_m^{r_i} \quad \text{for } i = 1, \dots, m-1.$$

Then, if $\alpha = (\alpha_1, \dots, \alpha_m) \in \mathbb{N}^m$, we consider the monomial $x^\alpha := x_1^{\alpha_1} \dots x_m^{\alpha_m}$ and we have

$$\begin{aligned} x^\alpha &= (y_1 + x_m^{r_1})^{\alpha_1} \dots (y_{m-1} + x_m^{r_{m-1}})^{\alpha_{m-1}} \cdot x_m^{\alpha_m} \\ &= x_m^{N(\alpha)} + \sum_{i=0}^{N(\alpha)-1} g_{\alpha,i} x_m^i \end{aligned}$$

where

$$N(\alpha) = \sum_{i=1}^{m-1} r_i \alpha_i + \alpha_m$$

and $g_{\alpha,i} \in k[y_1, \dots, y_{m-1}]$ for $i = 0, \dots, N(\alpha) - 1$.

Thus, if we write

$$f(x_1, \dots, x_m) = \sum_{\alpha \in \mathbb{N}^m} a_\alpha x^\alpha \quad \text{with } a_\alpha \in k$$

then

$$f(x_1, \dots, x_m) = \sum_{\alpha \in \mathbb{N}^m} a_\alpha \cdot (x_m^{N(\alpha)} + \sum_i g_{\alpha,i} x_m^i).$$

We want to choose $r_1, \dots, r_{m-1}$ so that the function

$$N: \{\alpha \in \mathbb{N}^m \mid a_\alpha \neq 0\} \rightarrow \mathbb{N} \quad \alpha \mapsto N(\alpha)$$

is injective, so that no cancellation among the top powers of x_m take place. For example, let r be a positive integer such that $r \geq \alpha_i$ for all $\alpha \in \mathbb{N}^m$ such that $a_\alpha \neq 0$ and let $r_i := r^{m-i}$. Then, it is easy to check that

$$N(\alpha) = r^{m-1}\alpha_1 + r^{m-2}\alpha_2 + \cdots + r\alpha_{m-1} + \alpha_m$$

satisfies our requirement.

Thus, if $N = \max\{N(\alpha) \mid a_\alpha \neq 0\}$, we have

$$f = ax_m^N + \sum_{i=0}^{N-1} g_i x_m^i$$

where $a \in k$ is non-zero and $g_i \in k[y_1, \dots, y_{m-1}]$.

Since $f(z_1, \dots, z_m) = 0$, it follows that z_m is integral over $k[y_1, \dots, y_{m-1}]$ and we can proceed as above. $\square$

Exercise 3.18. Let $A = \mathbb{Q}[x, y, z]/(xy - z^2)$. Find a polynomial extension B of $\mathbb{Q}$ such that A is finite over B .

3.4. Nullstellensatz theorem.

Lemma 3.19. Let A and B be integral domains such that $A \subset B$ and B is integral over A . Then A is a field if and only if B is a field.

Proof. Assume first that A is a field and let $b \in B$ be a non-zero element. Then b is integral over A and therefore there exist $a_0, \dots, a_{n-1} \in A$ such that

$$b^n + a_{n-1}b^{n-1} + \cdots + a_0 = 0.$$

We may assume that $a_0 \neq 0$ as otherwise we can divide by b and we can get an equation of lower degree. Define

$$c = -a_0^{-1} \cdot (b^{n-1} + a_{n-1}b^{n-2} + \cdots + a_1) \in B.$$

It is easy to check that $b \cdot c = 1$, i.e. c is the inverse of b . Thus, B is a field.

Assume now that B is a field and let $a \in A$ be a non-zero element. Then $c := a^{-1} \in B$ and, by assumption, c is integral over A . Thus, there exist $a_0, \dots, a_{n-1} \in A$ such that

$$c^n + a_{n-1}c^{n-1} + \cdots + a_0 = 0.$$

Thus, dividing by c^{n-1} , we have

$$c = -a_{n-1} - a_{n-2}a - \cdots - a_0a^{n-1} \in A.$$

It follows that A is a field. $\square$

Let k be a field and let $k \subset K$ be a field extension. Then we say that K is a **finite field extension** if K is a finite dimensional vector space over k , i.e. if K is a finitely generated module over k . Recall that if k is an algebraically closed field then any finite field extension of k coincides with k .

Theorem 3.20 (Weak Nullstellensatz). *Let k be a field and let $k \subset K$ be a field extension such that K is a finitely generated algebra over k . Then $k \subset K$ is a finite field extension.*

Moreover if k is algebraically closed then $k = K$.

Proof. By Noether normalisation theorem (cf. Theorem 3.17), there exist $a_1, \dots, a_m \in K$ algebraically independent and such that K is finite over $k[a_1, \dots, a_m]$. By Proposition 3.7, K is integral over $k[a_1, \dots, a_m]$ and since K is a field, Lemma 3.19 implies that also $k[a_1, \dots, a_m]$ is a field.

Since $a_1, \dots, a_m \in K$ are algebraically independent, we have that $k[a_1, \dots, a_m]$ is isomorphic to a polynomial ring of m variables and this is a field if and only if $m = 0$. Thus, K is finite over k and, in particular, if k is algebraically closed then $K = k$. $\square$

Corollary 3.21. *Let k be an algebraically closed field and let $\mathfrak{m} \subset k[x_1, \dots, x_n]$ be a maximal ideal. Then there exist $\alpha_1, \dots, \alpha_n \in k$ such that*

$$\mathfrak{m} = (x_1 - \alpha_1, \dots, x_n - \alpha_n).$$

Recall from Example 1.25 that the ideal $\mathfrak{m} = (x_1 - \alpha_1, \dots, x_n - \alpha_n) \subset k[x_1, \dots, x_n]$ is always maximal, even without assuming that k is algebraically closed.

Proof. As in Exercise 1.23, we have that $K := k[x_1, \dots, x_n]/\mathfrak{m}$ is a field. Moreover K is finitely generated over k , with generators $x_1, \dots, x_n$. Since k is algebraically closed, Theorem 3.20 implies that the natural inclusion $i: k \hookrightarrow K$ is an isomorphism. Let $\alpha_1, \dots, \alpha_n$ in k such that $i(\alpha_i) = [x_i] \in K$, for $i = 1, \dots, n$. Then it follows that $x_i - \alpha_i \in \mathfrak{m}$, i.e.

$$(x_1 - \alpha_1, \dots, x_n - \alpha_n) \subset \mathfrak{m}.$$

Since $(x_1 - \alpha_1, \dots, x_n - \alpha_n)$ is maximal, the equality holds. $\square$

Corollary 3.22. *Let k be an algebraically closed field and let $I \subset k[x_1, \dots, x_n]$ be a proper ideal. Then*

$$V(I) \neq \emptyset.$$

Proof. By Proposition 1.28, there exists a maximal ideal $\mathfrak{m} \subset k[x_1, \dots, x_n]$ such that $I \subset \mathfrak{m}$. By Corollary 3.21, there exist $\alpha_1, \dots, \alpha_n \in k$ such that

$$\mathfrak{m} = (x_1 - \alpha_1, \dots, x_n - \alpha_n).$$

Thus,

$$(\alpha_1, \dots, \alpha_n) \in V(\mathfrak{m}) \subset V(I)$$

and our claim follows. $\square$

Theorem 3.23 (Strong Nullstellensatz). *Let k be an algebraically closed field and let $I \subset k[x_1, \dots, x_n]$ be an ideal. Then*

$$I(V(I)) = \sqrt{I}.$$

Proof. Recall that by Lemma 1.39 we have that $\sqrt{I} \subset I(V(I))$.

Let $f \in I(V(I))$. Then $f(x) = 0$ for all $x \in V(I)$. We consider the ring extension

$$k[x_1, \dots, x_n] \subset k[x_1, \dots, x_n, y]$$

and the ideal

$$J = (I, yf - 1) \subset k[x_1, \dots, x_n, y].$$

Assume by contradiction that there exists $(a_1, \dots, a_n, b) \in V(J) \subset \mathbb{A}_k^{n+1}$ then if we denote $x := (a_1, \dots, a_n)$ we have that

$$x \in V(I) \quad \text{and} \quad b \cdot f(x) = 1.$$

This implies that $f(x) \neq 0$, a contradiction.

Thus, $V(J) = \emptyset$ and Corollary 3.22 implies that $J = k[x_1, \dots, x_n, y]$. Thus, there exist $g_0, \dots, g_k \in k[x_1, \dots, x_n, y]$, $h_1, \dots, h_k \in I$ such that

$$1 = \sum_{i=1}^k g_i \cdot h_i + g_0 \cdot (yf - 1).$$

It is easy to check that there exists a sufficiently large integer m such that, after multiplying by f^m on both sides, we get

$$f^m = \sum_{i=1}^k G_i(x_1, \dots, x_n, fy) \cdot h_i + G_0(x_1, \dots, x_n, fy) \cdot (fy - 1).$$

for some $G_0, \dots, G_m \in k[x_1, \dots, x_n, y]$. The equality holds true for all $(x_1, \dots, x_n, y) \in \mathbb{A}_k^{n+1}$. Thus, we may restrict to the locus where $yf(x_1, \dots, x_n) = 1$ and we get that

$$f^m = \sum_{i=1}^m G_i(x_1, \dots, x_m, 1) \cdot h_i \in I.$$

Thus, $f \in \sqrt{I}$ and our claim follows. $\square$

Recall by Proposition 1.44 that if R is a commutative ring and $I \subset R$ is an ideal then the radical $\sqrt{I}$ of I coincides with the intersection of all the prime ideals containing I . We now show that if R is a polynomial ring over an algebraically closed field, then it is enough to consider the intersection of all the maximal ideals containing I .

Proposition 3.24. *Let k be an algebraically closed field, let $R = k[x_1, \dots, x_n]$ and let $I \subset R$ be a proper ideal.*

Then

$$\sqrt{I} = \bigcap_{I \subset \mathfrak{m}} \mathfrak{m}$$

where the intersection runs over all the prime ideals $\mathfrak{m}$ containing I .

Proof. Since any maximal ideal is prime, Proposition 1.44 implies that $\sqrt{I} \subset \bigcap_{I \subset \mathfrak{m}} \mathfrak{m}$.

Let $f \in \bigcap_{I \subset \mathfrak{m}} \mathfrak{m}$ and let $a := (a_1, \dots, a_n) \in V(I)$. Then, as in Example 1.25, the ideal

$$\mathfrak{m}_a := I(\{a\}) = (x_1 - \alpha_1, \dots, x_n - \alpha_n) \subset k[x_1, \dots, x_n]$$

is maximal and Lemma 1.39 implies that

$$I \subset I(V(I)) \subset \mathfrak{m}_a.$$

Thus, $f \in \mathfrak{m}_a$, since f is contained in all the maximal ideals of $k[x_1, \dots, x_n]$ containing I . It follows that $f(a) = 0$ for all $a \in V(I)$. Therefore, $f \in I(V(I))$.

Theorem 3.23 implies that $f \in \sqrt{I}$ and our claim follows. $\square$

3.5. Ax theorem.

Theorem 3.25. *Let A be a finitely generated ring extension of $\mathbb{Z}$. Let $\mathfrak{m} \subset A$ be a maximal ideal. Then $A/\mathfrak{m}$ is a finite field.*

Proof. Let $\mathfrak{n} := \mathfrak{m} \cap \mathbb{Z}$. Then $\mathfrak{n}$ is a prime ideal of $\mathbb{Z}$. Since $1 \notin \mathfrak{n}$, we have that $\mathfrak{n} \neq \mathbb{Z}$. Thus, $\mathfrak{n} = (p)$ for some prime $p \in \mathbb{Z}$ or $\mathfrak{n} = (0)$.

Assume first that $\mathfrak{n} = (p)$. Then, by the assumption it follows that $A/\mathfrak{m}$ is a field which is a finitely generated $\mathbb{Z}/(p)$ -algebra. Thus, by Theorem 3.20, we have that $A/\mathfrak{m}$ is a finite extension of $\mathbb{Z}/(p) = \mathbb{Z}/p\mathbb{Z}$ and, in particular, it is finite.

Assume now that $\mathfrak{n} = (0)$. We will see that this case leads to a contradiction. We have that $A/\mathfrak{m}$ is a field that contains $\mathbb{Z}/(0) = \mathbb{Z}$. Hence $\mathbb{Q} \subset A/\mathfrak{m}$. It follows that $A/\mathfrak{m}$ is a finitely generated algebra over $\mathbb{Q}$ and by Theorem 3.20, we have that $A/\mathfrak{m}$ is a finite extension of $\mathbb{Q}$. Let $x_1, \dots, x_n \in A/\mathfrak{m}$ be generators of $A/\mathfrak{m}$ over $\mathbb{Q}$. Then there exist monic polynomials $f_1, \dots, f_n \in \mathbb{Q}[x]$ such that $f_i(x_i) = 0$ for $i = 1, \dots, n$. We may assume that $a_i \in \mathbb{Z}$ is the common denominator of the coefficients of f_i for each $i = 1, \dots, n$. Therefore $A/\mathfrak{m}$ is an integral extension of the ring $\mathbb{Z}[a_1^{-1}, \dots, a_n^{-1}]$. Since $A/\mathfrak{m}$ is a field, Lemma 3.19 implies that $\mathbb{Z}[a_1^{-1}, \dots, a_n^{-1}]$ is also a field. Thus, $\mathbb{Z}[a_1^{-1}, \dots, a_n^{-1}] = \mathbb{Q}$, which easily implies a contradiction. $\square$

Let k be a field. Recall from §1.3 that if $p_1, \dots, p_n \in k[x_1, \dots, x_n]$ are polynomials, then we have a regular mapping

$$p: \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n \quad (x_1, \dots, x_n) \mapsto (p_1(x_1, \dots, x_n), \dots, p_n(x_1, \dots, x_n)).$$

The following result is one of the many important applications of the Nullstellensatz.

Theorem 3.26 (Ax Theorem). *Let*

$$p: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$$

be an injective regular mapping.

Then p is a bijection.

Before proceeding with the proof, we first consider the case of a general field k . First note that if k is a finite field then $\mathbb{A}_k^n$ is finite and therefore any injective regular mapping $F: \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$ is automatically surjective.

We now consider the case of a general algebraically closed field k and we assume that

$$F := (f_1, \dots, f_n): \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$$

is a regular mapping, where $f_1, \dots, f_n \in k[x_1, \dots, x_n]$. We will need the following two remarks in the proof of Theorem [3.26](#):

- Let $I \subset k[x_1, \dots, x_n, y_1, \dots, y_n]$ be the ideal given by

$$I := (f_1(x_1, \dots, x_n) - f_1(y_1, \dots, y_n), \dots, f_n(x_1, \dots, x_n) - f_n(y_1, \dots, y_n)).$$

Assume that F is injective and let $(x_1, \dots, x_n, y_1, \dots, y_n) \in V(I)$. Then we have that

$$f_i(x_1, \dots, x_n) = f_i(y_1, \dots, y_n) \quad \text{for all } i = 1, \dots, n.$$

Thus, since F is injective, it follows that $x_i = y_i$ for all i , which implies that

$$x_i - y_i \in I(V(I)).$$

Since k is algebraically closed, the Nullstellensatz (cf. Theorem [3.23](#)) implies that

$$I(V(I)) = \sqrt{I}.$$

Thus, for each $i = 1, \dots, n$, there exist a positive integer ℓ_i and $q_{i,j} \in k[x_1, \dots, x_n, y_1, \dots, y_n]$ for $j = 1, \dots, n$ such that

$$(2) \quad (x_i - y_i)^{\ell_i} = \sum_{j=1}^n (f_j(x_1, \dots, x_n) - f_j(y_1, \dots, y_n)) \cdot q_{i,j}(x_1, \dots, x_n, y_1, \dots, y_n).$$

Conversely, if Equation [2](#) is satisfied then F is injective, even without the assumption of k being algebraically closed.

- Let $(a_1, \dots, a_n) \in \mathbb{A}_k^n$ such that

$$F(x_1, \dots, x_n) \neq (a_1, \dots, a_n) \quad \text{for all } (x_1, \dots, x_n) \in \mathbb{A}_k^n.$$

Then if we consider the ideal

$$J := (f_1(x_1, \dots, x_n) - a_1, \dots, f_n(x_1, \dots, x_n) - a_n) \subset k[x_1, \dots, x_n],$$

we have that $V(J) = \emptyset$. Since k is algebraically closed, Corollary [3.22](#) implies that the ideal is not proper, i.e. $1 \in J$ and, therefore, there exist $h_1, \dots, h_n \in k[x_1, \dots, x_n]$ such that

$$(3) \quad \sum_{i=1}^n (f_i - a_i) \cdot h_i = 1.$$

Conversely, if Equation [3](#) holds then $(a_1, \dots, a_n) \notin \text{Im} F$ even without assuming that k is algebraically closed. In particular, $F: \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$ is not surjective.

We can now proceed with the proof of Ax Theorem:

Proof of Theorem 3.26. We assume by contradiction that $p: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$ is injective but not surjective. Since $\mathbb{C}$ is algebraically closed, there exist:

- $\ell_1, \dots, \ell_n > 0$,
- $q_{i,j} \in \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n]$ for $i, j = 1, \dots, n$.
- $h_i \in \mathbb{C}[x_1, \dots, x_n]$ for $i = 1, \dots, n$, and
- $a_1, \dots, a_n \in \mathbb{C}$,

such that the Equations 2 and 3 hold. Let $A \subset \mathbb{C}$ be a finitely generated algebra over $\mathbb{Z}$ which is generated by all the coefficients of the polynomials $p_i, q_{i,j}, h_i$ for $i, j = 1, \dots, n$ and by the numbers $a_1, \dots, a_n$. In particular, we have that

$$p_i(a_1, \dots, a_n) \in A \quad \text{for all } a_1, \dots, a_n \in A.$$

Let $\mathfrak{m} \subset A$ be a maximal ideal. Then $k := A/\mathfrak{m}$ is a field (cf. Exercise 1.23). Fix $i = 1, \dots, n$. Since p_i is a polynomial and $\mathfrak{m} \subset A$ is an ideal, it follows easily that if $x, y \in A$ are such that $x - y \in \mathfrak{m}$ then $p_i(x) - p_i(y) \in \mathfrak{m}$. Thus, we have an induced regular mapping

$$F := \bar{p}: \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n.$$

Equations 2 and 3 continue to hold over k and hence F is injective but not surjective. On the other hand, Theorem 3.25 implies that k is a finite field, and therefore we have a contradiction. $\square$

Exercise 3.27. Find a regular mapping $p: \mathbb{A}_{\mathbb{Q}}^n \rightarrow \mathbb{A}_{\mathbb{Q}}^n$ which is injective but not surjective.

Exercise 3.28. Let $p: \mathbb{A}_{\mathbb{R}}^1 \rightarrow \mathbb{A}_{\mathbb{R}}^1$ be a regular mapping. Show that if it is injective then it is also surjective.

The same result holds more in general for any regular mapping $p: \mathbb{A}_{\mathbb{R}}^n \rightarrow \mathbb{A}_{\mathbb{R}}^n$, but the proof is much harder⁶

4. DIMENSION OF LOCAL RINGS

Definition 4.1. Let $(R, \mathfrak{m})$ be a local ring (cf. Definition 1.46). Let M be a finitely generated module over R . We define $\mu(M)$ to be the minimum number of generators of M over R .

Proposition 4.2. Let $(R, \mathfrak{m})$ be a local ring and let $k = R/\mathfrak{m}$ be its residue field. Let M be a finitely generated module over R . Then $M/\mathfrak{m}M$ is a vector space over k of dimension $\mu(M)$.

⁶A. S. Białynicki-Birula and M. Rosenlicht, Injective morphisms of real algebraic varieties, Proc. Amer. Math. Soc. **13** (1962), 200–203; MR0140516

Proof. It is easy to check that $M/\mathfrak{m}M$ is a vector space over k and that if $x_1, \dots, x_n$ are generators of M over R then $[x_1], \dots, [x_n]$ are generators of $M/\mathfrak{m}M$ over k . Thus,

$$\dim_k M/\mathfrak{m}M \leq \mu(M).$$

Conversely, assume that $x_1, \dots, x_n \in M$ are such that $[x_1], \dots, [x_n]$ are generators of $M/\mathfrak{m}M$ over k . Let $N \subset M$ be the submodule generated by $x_1, \dots, x_n$. Consider the R -module

$$\overline{M} := M/N.$$

We want to show that $\overline{M} = 0$. We first show that $\mathfrak{m}\overline{M} = \overline{M}$. Let $x \in M$. Then there exist $r_1, \dots, r_n \in R$ such that

$$[x] = \sum_{i=1}^n [r_i][x_i] \quad \text{in } M/\mathfrak{m}M.$$

Thus,

$$x - \sum_{i=1}^n r_i x_i \in \mathfrak{m}M$$

and, in particular, the image of x in $\overline{M}$ is contained in $\mathfrak{m}\overline{M}$. Thus, $\mathfrak{m}\overline{M} = \overline{M}$.

Note that since $(R, \mathfrak{m})$ is a local ring, we have that $\mathcal{J}(R) = \mathfrak{m}$. Therefore, by Corollary 2.14, we have that $\overline{M} = 0$, which implies that $N = M$. It follows that

$$\dim_k M/\mathfrak{m}M \geq \mu(M).$$

Thus, the equality holds. $\square$

Definition 4.3. Let $(R, \mathfrak{m})$ be a local Noetherian ring with residue field $k := R/\mathfrak{m}$. Then we define

$$\mu(R) := \dim_k \mathfrak{m}/\mathfrak{m}^2.$$

R is called **regular** if $\mu(R) = \dim R$.

Note that the definition is compatible with Proposition 4.2. Indeed, since R is Noetherian, we have that $M := \mathfrak{m}$ is a finitely generated R -module and $\mathfrak{m}M = \mathfrak{m}^2$.

Definition 4.4. Let R be a commutative ring. The **Spectrum** of R is the set

$$\text{Spec}(R) := \{\mathfrak{p} \subset R \mid \mathfrak{p} \text{ is a prime ideal of } R\}.$$

Exercise 4.5. Let R be a commutative ring, let $\mathfrak{p} \subset R$ be a prime ideal and let $R_{\mathfrak{p}}$ be the localisation of R at $\mathfrak{p}$ (cf. Section 1.5). Then there exists a bijection between $\text{Spec}(R_{\mathfrak{p}})$ and the set

$$\{\mathfrak{q} \in \text{Spec} R \mid \mathfrak{q} \subset \mathfrak{p}\}.$$

In particular, we have

$$\dim R_{\mathfrak{p}} = h(\mathfrak{p})$$

(cf. Definition 2.24).

4.1. Krull's principal ideal theorem.

Theorem 4.6 (Krull's Principal Ideal Theorem). *Let R be a Noetherian ring and let $a \in R$ be a non-unit. Let $\mathfrak{p}$ be a prime ideal which is minimal among all the prime ideals such that $a \in \mathfrak{p}$. Then*

$$h(\mathfrak{p}) \leq 1.$$

Proof. Let $R_{\mathfrak{p}}$ be the localisation of R at $\mathfrak{p}$. Then $R_{\mathfrak{p}}$ is Noetherian and the maximal ideal of $R_{\mathfrak{p}}$ is $\mathfrak{m}_{\mathfrak{p}} := \mathfrak{p}R_{\mathfrak{p}}$. We have that $\mathfrak{m}_{\mathfrak{p}}$ is a minimal prime ideal containing $\frac{a}{1} \in R_{\mathfrak{p}}$. Since $\mathfrak{m}_{\mathfrak{p}}$ is maximal, we have that $\dim R_{\mathfrak{p}} = h(\mathfrak{m}_{\mathfrak{p}})$ and by Exercise 4.5, we have that $\dim R_{\mathfrak{p}} = h(\mathfrak{p})$. Therefore, it is enough to show that $h(\mathfrak{m}_{\mathfrak{p}}) \leq 1$. Thus, after replacing R by $R_{\mathfrak{p}}$ we may assume that R is a local ring with maximal ideal $\mathfrak{m} := \mathfrak{p}$.

Assume that $h(\mathfrak{p}) > 0$, as otherwise there is nothing to show. Let $\mathfrak{q} \subsetneq \mathfrak{p}$ be a prime ideal. In particular, $a \notin \mathfrak{q}$. We want to show that $h(\mathfrak{q}) = 0$. Let $R_{\mathfrak{q}}$ be the localisation of R at $\mathfrak{q}$, with maximal ideal $\mathfrak{m}_{\mathfrak{q}} := \mathfrak{q}R_{\mathfrak{q}}$. We first show that

$$\mathfrak{m}_{\mathfrak{q}} \subset \sqrt{(0)} \quad \text{in } R_{\mathfrak{q}}.$$

To this end, we define the ideal

$$\mathfrak{q}^{(n)} := \{r \in R \mid rt \in \mathfrak{q}^n \text{ for some } t \in R \setminus \mathfrak{q}\} \subset R$$

Note that if $\varphi: R \rightarrow R_{\mathfrak{q}}$ is the natural map, then

$$\mathfrak{q}^{(n)} = \varphi^{-1}(\mathfrak{q}^n R_{\mathfrak{q}}).$$

We have a descending chain of ideals in R

$$\mathfrak{q} = \mathfrak{q}^{(1)} \supset \mathfrak{q}^{(2)} \supset \dots$$

which induces the chain of ideals in R

$$\mathfrak{p} \supset (a) + \mathfrak{q}^{(1)} \supset (a) + \mathfrak{q}^{(2)} \supset \dots$$

which, in turn, induces the chain of ideals in $\overline{R} := R/(a)$

$$\overline{\mathfrak{p}} \supset \overline{\mathfrak{q}}^{(1)} \supset \overline{\mathfrak{q}}^{(2)} \supset \dots$$

Since $\mathfrak{p}$ is the only maximal ideal of R and, at the same time, it is a minimal prime ideal of R containing a , it follows easily that $\overline{\mathfrak{p}}$ is the only prime ideal of $\overline{R}$. Thus, $\dim \overline{R} = 0$ and, since $\overline{R}$ is Noetherian (cf. Exercise 2.22), by Theorem 2.42 we have that $\overline{R}$ is also Artinian. Thus, there exists a positive integer n such that

$$\overline{\mathfrak{q}}^{(i)} = \overline{\mathfrak{q}}^{(n)} \quad \text{for all } i \geq n$$

and, therefore,

$$(a) + \mathfrak{q}^{(i)} = (a) + \mathfrak{q}^{(n)} \quad \text{for all } i \geq n.$$

⁷ $\mathfrak{q}^{(n)}$ is called the n^{th} -symbolic power of $\mathfrak{q}$

Thus, if $x \in \mathfrak{q}^{(n)}$, there exists $r \in R$ and $y \in \mathfrak{q}^{(n+1)}$ such that $x = ra + y$ and, therefore, $ra = x - y \in \mathfrak{q}^{(n)}$. Since $\mathfrak{q}^{(n)} = \varphi^{-1}(\mathfrak{q}^n R_{\mathfrak{q}})$, it follows that

$$\frac{ra}{1} \in \mathfrak{q}^n R_{\mathfrak{q}}.$$

Since $a \notin \mathfrak{q}$, we have that $\frac{1}{a} \in R_{\mathfrak{q}}$ and, therefore, $\frac{r}{1} \in \mathfrak{q}^{(n)} R_{\mathfrak{q}}$. Since $\mathfrak{q}^{(n)} = \varphi^{-1}(\mathfrak{q}^n R_{\mathfrak{q}})$, it follows that $r \in \mathfrak{q}^{(n)}$. Thus, since $x = ra + y$ and $a \in \mathfrak{p}$, we have proven that

$$\mathfrak{q}^{(n)} = \mathfrak{p}\mathfrak{q}^{(n)} + \mathfrak{q}^{(n+1)}.$$

Since $(R, \mathfrak{p})$ is a local ring, by Corollary 2.14 it follows that $\mathfrak{q}^{(n)} = \mathfrak{q}^{(n+1)}$ and, in particular,

$$\mathfrak{q}^n R_{\mathfrak{q}} = \mathfrak{q}^{n+1} R_{\mathfrak{q}} = (\mathfrak{q} R_{\mathfrak{q}}) \cdot (\mathfrak{q}^n R_{\mathfrak{q}}).$$

By applying Corollary 2.14 again, it follows that $\mathfrak{q}^n R_{\mathfrak{q}} = 0$, which implies that

$$\mathfrak{m}_{\mathfrak{q}} = \mathfrak{q} R_{\mathfrak{q}} \subset \sqrt{(0)},$$

i.e. $\mathfrak{m}_{\mathfrak{q}}$ is nilpotent, as claimed.

By Proposition 1.44, it follows that $\sqrt{(0)}$ is the intersection of all the prime ideals of $R_{\mathfrak{q}}$. Since $\mathfrak{m}_{\mathfrak{q}}$ is maximal in $R_{\mathfrak{q}}$, it follows that $\mathfrak{m}_{\mathfrak{q}}$ is the only prime ideal of $R_{\mathfrak{q}}$. Thus, $h(\mathfrak{q}) = \dim R_{\mathfrak{q}} = 0$ and, therefore, $h(\mathfrak{p}) \leq 1$. $\square$

Exercise 4.7. Let R be a Noetherian ring. Let

$$\mathfrak{p} \subsetneq \mathfrak{r}_0 \subsetneq \mathfrak{q} \subset R$$

be prime ideals. Show that there are infinitely many prime ideals $\mathfrak{r} \subset R$ such that

$$\mathfrak{p} \subsetneq \mathfrak{r} \subsetneq \mathfrak{q} \subset R.$$

4.2. Krull's height theorem.

Theorem 4.8 (Krull's height Theorem). Let R be a Noetherian ring, let $a_1, \dots, a_r \in R$ be such that $I := (a_1, \dots, a_r) \subset R$ is a proper ideal. Let $\mathfrak{p}$ be a prime ideal which is minimal among all the prime ideals such that $a_1, \dots, a_r \in \mathfrak{p}$.

Then

$$h(\mathfrak{p}) \leq r.$$

Proof. We proceed by induction on r . Note that the case $r = 1$ follows from Theorem 4.6. As in the proof of Theorem 4.6, we may assume that $(R, \mathfrak{p})$ is a local ring with maximal ideal $\mathfrak{p}$.

Let $\mathfrak{p}_1 \subsetneq \mathfrak{p}$ be a prime ideal which is maximal with respect to all the prime ideals properly contained in $\mathfrak{p}$. Note that

$$h(\mathfrak{p}_1) = h(\mathfrak{p}) - 1.$$

Since $\mathfrak{p}$ is a minimal prime ideal containing I , we have that I is not contained in $\mathfrak{p}_1$ and, we may assume that $a_1 \notin \mathfrak{p}_1$. Let

$$J := (a_1) + \mathfrak{p}_1 \subset \mathfrak{p}.$$

Since $h(\mathfrak{p}_1) = h(\mathfrak{p}) - 1$, we have that $\mathfrak{p}$ is the minimal prime ideal containing J .

We want to show that $\sqrt{J} = \mathfrak{p}$. By Theorem 2.40, there exist $\mathfrak{q}_1, \dots, \mathfrak{q}_m \subset R$ prime ideals such that

$$\sqrt{J} = \bigcap_{i=1}^m \mathfrak{q}_i.$$

Since $\mathfrak{p}$ is the only maximal ideal in R , we have that $J \subset \mathfrak{q}_i \subset \mathfrak{p}$ for all $i = 1, \dots, m$ and since $\mathfrak{p}$ is minimal among the prime ideals containing J , we have that $\mathfrak{p} = \mathfrak{q}_i$ for all $i = 1, \dots, m$. Thus, $\sqrt{J} = \mathfrak{p}$ and, in particular, there exists a positive integer n such that

$$a_i^n \in J \quad \text{for all } i = 1, \dots, r.$$

Hence, for all $i = 1, \dots, r$, there exist $x_i \in R$ and $y_i \in \mathfrak{p}_1$ such that

$$a_i^n = a_1 x_i + y_i.$$

Let

$$I_1 := (y_2, \dots, y_r) \subset \mathfrak{p}_1 \subset R.$$

We want to show that $\mathfrak{p}_1$ is a minimal prime ideal containing I_1 . Let $S = R/I_1$ and let $\bar{\mathfrak{p}}$ and $\bar{\mathfrak{p}}_1$ be the image of $\mathfrak{p}$ and $\mathfrak{p}_1$ in S respectively and let $\bar{a}_1 := [a_1] \in S$. Then

$$(0) \subsetneq (\bar{a}_1) \subset \bar{\mathfrak{p}} \subset S.$$

We first show that $\bar{\mathfrak{p}} = \sqrt{(\bar{a}_1)}$. We have

$$(a_1, y_2, \dots, y_r) = (a_1, a_2^n - a_1 x_2, a_3^n - a_1 x_3, \dots, a_r^n - a_1 x_r) = (a_1, a_2^n, a_3^n, \dots, a_r^n).$$

Since $\mathfrak{p}$ is the minimal ideal containing $I = (a_1, \dots, a_n)$, similarly as above, we have that $\mathfrak{p} = \sqrt{I}$. Thus, for every $x \in \mathfrak{p}$, there exists a positive integer s such that $x^s \in I$ and, therefore,

$$x^{srn} \in (a_1^n, a_2^n, a_3^n, \dots, a_r^n) \subset (a_1, y_2, \dots, y_r).$$

Thus,

$$\sqrt{(a_1, y_2, \dots, y_r)} = \mathfrak{p}$$

which implies that $\bar{\mathfrak{p}} = \sqrt{(\bar{a}_1)}$, as claimed. By Theorem 4.6 it follows that $h(\bar{\mathfrak{p}}) \leq 1$ and, therefore, $h(\bar{\mathfrak{p}}_1) = 0$. Thus, $\mathfrak{p}_1$ is the minimal prime ideal containing I_1 . By induction, it follows that

$$h(\mathfrak{p}) = h(\mathfrak{p}_1) + 1 \leq (r - 1) + 1 = r,$$

as claimed. □

Corollary 4.9. *Let $(R, \mathfrak{m})$ be a local Noetherian ring. Then*

$$\dim R \leq \mu(R).$$

In particular, R has finite dimension.

Proof. Recall that, as in Definition 4.3 we have that

$$\mu(R) := \dim_k \mathfrak{m}/\mathfrak{m}^2$$

and by Proposition 4.2 we have that $\mu(R) = \mu(\mathfrak{m})$, that is the minimum number of generators of $\mathfrak{m}$ as a R -module. Since R is Noetherian, we have that $\mu(R) < \infty$.

Let $r = \mu(R)$ and let $x_1, \dots, x_r \in \mathfrak{m}$ be generators of $\mathfrak{m}$. Then $\mathfrak{m}$ is the minimal prime ideal containing $x_1, \dots, x_r$ and, by Theorem 4.8 we have that

$$\dim(R) = h(\mathfrak{m}) \leq r = \mu(R)$$

as claimed. $\square$

Corollary 4.10. *Let k be an algebraically closed field.⁸ Then*

$$\dim k[x_1, \dots, x_n] = n.$$

Proof. We showed (cf. Exercise 2.25) that $\dim k[x_1, \dots, x_n] \geq n$. Let

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \mathfrak{p}_2 \dots \subsetneq \mathfrak{p}_n \subsetneq k[x_1, \dots, x_n]$$

be a chain of prime ideals with $\mathfrak{p}_n$ maximal. By Corollary 3.21 there exist $\alpha_1, \dots, \alpha_n \in k$ such that

$$\mathfrak{p}_n = (x_1 - \alpha_1, \dots, x_n - \alpha_n).$$

Thus, $\mathfrak{p}_n$ is the minimal prime ideal containing $x_1 - \alpha_1, \dots, x_n - \alpha_n$ and, by Theorem 4.8 we have that

$$\dim k[x_1, \dots, x_n] = h(\mathfrak{p}_n) \leq n.$$

Thus, our claim follows. $\square$

5. DISCRETE VALUATION RINGS

Definition 5.1. *Let K be a field. A **discrete valuation** of K is a surjective map*

$$\nu: K \setminus \{0\} \rightarrow \mathbb{Z}$$

such that, for all $x, y \in K \setminus \{0\}$, we have

- (1) $\nu(xy) = \nu(x) + \nu(y)$;
- (2) $\nu(x \pm y) \geq \min\{\nu(x), \nu(y)\}$

We usually denote $\nu(0) = \infty$ so that (1) and (2) continue to hold.

Proposition 5.2. *Let K be a field and let $\nu: K \setminus \{0\} \rightarrow \mathbb{Z}$ be a discrete valuation of k . Then*

- (1) $\nu(x^{-1}) = -\nu(x)$ for all $x \in K \setminus \{0\}$.

⁸The same result follows for any field, without the assumption of being algebraically closed, but we omit the proof here.

(2) the set

$$R_\nu := \{x \in K \mid \nu(x) \geq 0\}$$

is a local ring, with maximal ideal

$$\mathfrak{m}_\nu := \{x \in K \mid \nu(x) > 0\} \subset R_\nu.$$

In particular, $x \in R_\nu$ is a unit, if and only if $\nu(x) = 0$.

- (3) if $t \in K$ is such that $\nu(t) = 1$ then $\mathfrak{m}_\nu = (t)$.
- (4) if $I \subset R_\nu$ is an ideal then there exists an integer $n \geq 0$ such that $I = (t^n)$. In particular, R_ν is Noetherian and $\dim R_\nu = 1$.
- (5) Let $k_\nu := R_\nu/\mathfrak{m}_\nu$. Then $\dim_{k_\nu} \mathfrak{m}_\nu/\mathfrak{m}_\nu^2 = 1$ and, in particular, R_ν is regular (cf. Definition 4.3).

The local ring $(R_\nu, \mathfrak{m}_\nu)$ is called a **discrete valuation ring** (DVR). A generator t of $\mathfrak{m}_\nu$ is called a **parameter** or **uniformiser** of R .

Proof. (1) and (2) are easy exercises.

We now prove (3). If $x \in \mathfrak{m}_\nu$ then $\nu(x) = n \in \mathbb{Z}_{>0}$. Let

$$u := \frac{x}{t^n} \in k.$$

Then $\nu(u) = \nu(ut^n) - \nu(t^n) = 0$ and, therefore, $u \in R_\nu$. Thus, $x = ut^n \in (t)$ and (3) follows.

To prove (4), it is enough to take

$$n = \min\{\nu(x) \mid x \in I\}.$$

Finally, since $\mathfrak{m}_\nu/\mathfrak{m}_\nu^2 \neq 0$, Proposition 4.2 implies (5). $\square$

Example 5.3. Let $k = \mathbb{Q}$ and fix a prime number p . Then any number $x \in k \setminus \{0\}$ can be written as

$$x = p^m \cdot \frac{a}{b} \quad \text{where } m \in \mathbb{Z} \quad \text{and} \quad \gcd(a, p) = \gcd(b, p) = 1.$$

Note that $m \in \mathbb{Z}$ is uniquely defined by x . Thus, we can define

$$v_p(x) := m.$$

The DVR of v_p is $\mathbb{Z}_{(p)}$ (cf. Example 1.50).

Lemma 5.4. Let R be a Noetherian integral domain and let $t \in R$ be a non-unit. Then

$$\bigcap_{n \geq 1} (t^n) = (0).$$

Proof. We may assume that $t \neq 0$ as, otherwise, there is nothing to prove. Assume, by contradiction that there exists

$$0 \neq x \in \bigcap_{n \geq 1} (t^n).$$

Then, for each $n \geq 1$, there exists $x_n \in R$ such that

$$x = x_n t^n.$$

Thus, $x_{n-1} = x_n \cdot t$ and, in particular,

$$(x) \subset (x_1) \subset (x_2) \subset \dots$$

Since R is Noetherian, there exists a positive integer m such that $(x_m) = (x_{m+1})$, which implies that

$$x_{m+1} = ax_m \quad \text{for some } a \in R.$$

Thus,

$$x = x_{m+1} t^{m+1} = ax_m t^{m+1} = atx.$$

Since R is an integral domain, it follows that $at = 1$, i.e. t is a unit, a contradiction. $\square$

Lemma 5.5. *Let $(R, \mathfrak{m})$ be a local ring, such that R is an integral domain and $\mathfrak{m} = (t)$ for some $t \in R$. Assume that*

$$\bigcap_{n \geq 1} (t^n) = 0$$

and let $k := \text{Frac} R$. Then

- (1) *If $x \in R$ is non-zero then $x = t^n u$ for some integer $n \geq 0$ and unit $u \in R$.*
- (2) *Define $\nu(x) := n$ and if $\frac{x}{y} \in k$, define*

$$\nu(x/y) := \nu(x) - \nu(y).$$

Then ν is a discrete valuation of k and $(R, \mathfrak{m})$ is its discrete valuation ring.

Proof. We first prove (1). If $x \in R$ is a unit, then we may write $x = t^0 \cdot u$ where $u = x$.

Assume now that x is not a unit. Since $\mathfrak{m}$ is the unique maximal ideal of R , by Proposition 1.28, it follows that $x \in \mathfrak{m} = (t)$. Since $\bigcap_{n \geq 1} (t^n) = 0$, there exists a positive integer n such that

$$x \in (t^n) \setminus (t^{n+1}).$$

Thus, $x = t^n \cdot u$ and $u \notin \mathfrak{m}$. Thus, u is a unit and (1) follows.

We now prove (2). Let $z \in k$. Then there exists $n \in \mathbb{Z}$ such that

$$z = t^n u \quad \text{where } u \in R \text{ is a unit.}$$

In particular $\nu(z) = n$. It is the easy to show that ν is a discrete valuation of k and $(R, \mathfrak{m})$ is its DVR. $\square$

Lemma 5.6. *Let R be a Noetherian ring and let M be a non-zero R -module. Then there exists $0 \neq x \in M$ such that $\text{ann}(Rx)$ is a prime ideal of R (cf. Exercise 1.59).*

Proof. Consider the set

$$P := \{\text{ann}(Rx) \mid 0 \neq x \in M\}.$$

Since R is Noetherian, by Zorn's Lemma (cf. Theorem 1.27), there exists $x \in M$ such that $\mathfrak{p} := \text{ann}(Rx)$ is maximal in P . We want to show that $\mathfrak{p} \subset R$ is a prime ideal. Assume by contradiction that there exist $a, b \in R$ such that $a, b \notin \mathfrak{p}$ and $ab \in \mathfrak{p}$. Then

$$a \cdot x \neq 0 \quad b \cdot x \neq 0 \quad \text{and} \quad ab \cdot x = 0.$$

In particular,

$$\text{ann}(Rx) \subset \text{ann}(R(a \cdot x)) \quad \text{and} \quad b \in \text{ann}(R(a \cdot x)) \setminus \text{ann}(Rx),$$

which contradicts the fact that $\mathfrak{p}$ is maximal in P . $\square$

Theorem 5.7. *Let $(R, \mathfrak{m})$ be a local ring.*

Then $(R, \mathfrak{m})$ is a DVR if and only if R is Noetherian, normal and

$$\text{Spec} R = \{(0), \mathfrak{m}\}.$$

Proof. Assume first that $(R, \mathfrak{m})$ is a DVR. The Proposition 5.2 immediately implies that R is Noetherian and $\text{Spec} R = \{(0), \mathfrak{m}\}$. Proposition 5.2 also implies that R is a PID and, in particular, a UFD (cf. Example 1.18). Thus, R is normal (cf. Exercise 3.12).

Assume now that R is Noetherian, normal and $\text{Spec} R = \{(0), \mathfrak{m}\}$. Let $k = \text{Frac}(R)$. Since $\mathfrak{m} \neq 0$, Proposition 2.14 implies that $\mathfrak{m} \neq \mathfrak{m}^2$. Let $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. We want to show that $\mathfrak{m} = (x)$. Consider the R -module $M := \mathfrak{m}/(x)$ and assume by contradiction that $M \neq 0$. Then by Lemma 5.6 there exists $y \in \mathfrak{m} \setminus (x)$ such that if $\bar{y} \in M$ is the projection of y in M then $\text{ann}(R\bar{y})$ is a prime ideal of R . Since $\mathfrak{m} \subset R$ is the only non zero prime ideal of R , it follows that $\text{ann}(R\bar{y}) = \mathfrak{m}$ and, therefore, $y \cdot \mathfrak{m} \subset (x)$.

Let

$$z := \frac{y}{x} \in k \setminus R.$$

Since $y \cdot \mathfrak{m} \subset (x)$, we have that $z \cdot \mathfrak{m} \subset R$. Assume that $z \cdot \mathfrak{m} = R$. Then there exists $w \in \mathfrak{m}$ such that $wz = 1$ and, therefore, we have $x = wy \in \mathfrak{m}^2$, a contradiction. Thus, $z \cdot \mathfrak{m} \neq R$ and since $\mathfrak{m}$ is the only maximal ideal in R , we have that $z \cdot \mathfrak{m} \subset \mathfrak{m}$.

We now want to show that z is integral over R . Consider the R -linear map

$$\varphi: \mathfrak{m} \rightarrow \mathfrak{m} \quad t \mapsto z \cdot t.$$

By Cayley-Hamilton's Theorem (cf. Theorem 2.11), here exist $a_0, \dots, a_{n-1} \in R$ such that

$$\varphi^n(t) + a_{n-1} \cdot \varphi^{n-1}(t) + \dots + a_1 \cdot \varphi(t) + a_0 \cdot t = 0$$

for all $t \in \mathfrak{m}$. Thus,

$$(z^n + a_{n-1} \cdot z^{n-1} + \dots + a_1 \cdot z + a_0) \cdot t = 0$$

for all $t \in \mathfrak{m}$. Since R is an integral domain, we have that z is integral over R , as claimed.

Since R is normal, it follows that $\frac{y}{x} = z \in R$. Thus, $y \in (x)$, which contradicts our assumption on y . Thus, $M = 0$ which implies that $\mathfrak{m} = (x)$.

Finally, Lemma 5.4 and Lemma 5.5 imply that $(R, \mathfrak{m})$ is a DVR, as claimed. $\square$

6. COMPLETION OF A FIELD

Let K be a field and $\nu : K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation. Let $r \in \mathbb{R}$ with $r > 1$. We set

$$|x|_\nu := r^{-\nu(x)}, \text{ for } x \in K.$$

We can verify that $|\cdot| : K \rightarrow \mathbb{R}_{\geq 0}$ is a norm on K . Namely, for all $x, y \in K$, the following properties are satisfied:

- (1) $|xy|_\nu = |x|_\nu |y|_\nu$
- (2) $|x + y|_\nu \leq \max\{|x|_\nu, |y|_\nu\}$; and
- (3) $|x|_\nu = 0 \iff x = 0$.

The second property is called the **non-archimedean property**. Note that, in particular, it implies the triangular inequality $|x + y|_\nu \leq |x|_\nu + |y|_\nu$. Hence, we can define a distance function $d(x, y) = |x - y|_\nu$ and in turn equip K with a topology with basic open sets

$$B_{x_0, \epsilon} := \{x \in K : |x - x_0|_\nu < \epsilon\}.$$

Note that the topology does not depend on the choice of $r > 1$. We next show that in non-archimedean geometry every triangle is isosceles:

Lemma 6.1. *If $|x|_\nu < |y|_\nu$ then $|x - y|_\nu = |y|_\nu$.*

Proof. We have $|x - y|_\nu \leq \max\{|x|_\nu, |y|_\nu\} = |y|_\nu$ since $|x|_\nu < |y|_\nu$. On the other hand, $y = (y - x) + x$, hence $|y|_\nu \leq \max\{|x - y|_\nu, |x|_\nu\} = |x - y|_\nu$ since $|x|_\nu < |y|_\nu$. $\square$

Definition 6.2. *Let K be a field and let $\nu : K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation. A sequence $x_n \in K$, with $n \geq 1$, is a **Cauchy sequence** if for any $\epsilon > 0$ there exists $n_0 \geq 1$ such that $|x_n - x_m|_\nu < \epsilon$ for all $n, m \geq n_0$.*

This is equivalent to the condition that for all $N > 0$ there exists $n_0 \geq 1$ such that $\nu(x_n - x_m) > N$ for all $n, m \geq n_0$.

*We say that the sequence $(x_n)_{n \geq 1}$ converges to a **limit** $x \in K$ if for all $\epsilon > 0$ there exists $n_0 \geq 0$ such that $|x_n - x|_\nu < \epsilon$ for all $n \geq n_0$. Or equivalently, for all $N > 0$ there exists $n_0 \geq 0$ such that $\nu(x_n - x) > N$ for all $n \geq n_0$.*

*The field K is called **complete** with respect to the valuation ν if every Cauchy sequence has a limit.*

Definition 6.3. *Let K be a field and $\nu : K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation. We say that a field $\hat{K}_\nu \supset K$ is a **completion** of K if the valuation ν extends to $\hat{K}_\nu$, it is complete and it is the closure of K with respect to the topology induced by ν .*

Given a field K with a valuation ν , there is a canonical construction of a completion of K_ν . We outline the construction. First, we set

$$\mathcal{C}_\nu := \{(x_n)_{n \geq 1} : x_n \in K, (x_n) \text{ is a Cauchy sequence}\},$$

and

$$\mathcal{I}_\nu := \{(x_n)_{n \geq 1} \in \mathcal{C}_\nu : \lim_{n \rightarrow \infty} x_n = 0\}.$$

Note that $\mathcal{C}_\nu$ is a commutative ring with

- (1) $(x_n) \pm (y_n) = (x_n \pm y_n)$
- (2) $(x_n) \cdot (y_n) = (x_n y_n)$;
- (3) $0 = (0)_{n \geq 1}$; and
- (4) $1 = (1)_{n \geq 1}$.

Moreover, $\mathcal{I}_\nu$ is a maximal ideal in $\mathcal{C}_\nu$, hence $\hat{K}_\nu := \mathcal{C}_\nu / \mathcal{I}_\nu$ is a field (cf. Exercise 1.23). We have that the map $K \rightarrow \hat{K}_\nu$ given by $x \rightarrow (x)_{n \geq 1}$ is an embedding, hence $K \subset \hat{K}_\nu$ is a field extension.

Lemma 6.4. *Let K be a field and $\nu : K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation. Let $0 \neq \alpha \in \hat{K}_\nu$ represented by the class of a sequence $(x_n)_{n \geq 1} \in \mathcal{C}_\nu / \mathcal{I}_\nu$. Then there exists $n_0 \geq 1$ such that $\nu(x_n) = \nu(x_{n_0})$ for all $n \geq n_0$. We define $\hat{\nu} : \hat{K}_\nu \rightarrow \mathbb{Z} \cup \{\infty\}$ by letting $\hat{\nu}(\alpha) = \lim_{n \rightarrow \infty} \nu(x_n) = \nu(x_{n_0})$. Then, $\hat{\nu}$ is a valuation extending ν .*

Proof. Since $\alpha \neq 0$, $(x_n) \notin \mathcal{I}_\nu$. Hence, there exists $c > 0$ such that for all $n \geq 1$, there exists $i_n > n$ with $\nu(x_{i_n}) \leq c$. On the other hand, (x_n) is a Cauchy sequence and, therefore, there exists $N \geq 1$ such that $\nu(x_n - x_m) > c$ for all $n, m \geq N$. Let $n_0 = i_N > N$. Then, for all $n \geq n_0$, we have

$$\nu(x_n) = \nu(x_n - x_{n_0} + x_{n_0}) \geq \min\{\nu(x_n - x_{n_0}), \nu(x_{n_0})\} = \nu(x_{n_0})$$

and

$$\nu(x_{n_0}) = \nu(x_{n_0} - x_n + x_n) \geq \min\{\nu(x_{n_0} - x_n), \nu(x_n)\} = \nu(x_n).$$

Hence, $\nu(x_n) = \nu(x_{n_0})$ for all $n \geq n_0$. We define $\hat{\nu}(\alpha) := \nu(x_{n_0})$.

It is an easy exercise to check that $\hat{\nu}$ is well-defined and a valuation. □

Thus, giving a field K with a discrete valuation $\nu : K \rightarrow \mathbb{Z} \cup \{\infty\}$, we have constructed a field extension $K \subset \hat{K}_\nu$ and a valuation $\hat{\nu} : \hat{K}_\nu \rightarrow \mathbb{Z} \cup \{\infty\}$ such that all the Cauchy sequences in $\hat{K}_\nu$ converge. In particular, if $\alpha \in \hat{K}_\nu$ is represented by $(x_n)_{n \geq 1} \in \mathcal{C}_\nu / \mathcal{I}_\nu$ then $\lim_{n \rightarrow \infty} x_n = \alpha$.

As in Proposition 5.2, let

$$\hat{R}_\nu := \{x \in \hat{K}_\nu : \hat{\nu}(x) \geq 0\}$$

be the corresponding discrete valuation ring in $\hat{K}_\nu$. Let $t \in R_\nu$ such that $\nu(t) = 1$. Proposition 5.2 implies that $\mathfrak{m}_\nu = (t)$ is the maximal ideal of R_ν . Since $\hat{\nu}|_{R_\nu} = \nu$, we have $\hat{\nu}(t) = 1$ and $\hat{\mathfrak{m}}_\nu = (t)$ is the unique maximal ideal of $\hat{R}_\nu$. Thus, we have a diagram

$$\begin{array}{ccccc} K_\nu & \supset & R_\nu & \supset & \mathfrak{m}_\nu \\ \cap & & \cap & & \cap \\ \hat{K}_\nu & \supset & \hat{R}_\nu & \supset & \hat{\mathfrak{m}}_\nu \end{array}$$

Exercise 6.5. *Let k be a field. Show that the completion of the discrete valuation ring $k[x]_{(x)}$ is isomorphic to the formal power series ring $k[[x]]$.*

Let K be a field and let $\nu : K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation. Let $(x_n)_{n \geq 1}$ be a sequence in K . If the sequence of partial sums $s_N := \sum_{n=1}^N x_n$ converges, then we will call the limit to be the **infinite sum** $\sum_{n \geq 1} x_n$. In a complete discrete valuation ring $\hat{R}_\nu$, the sum converges if and only if $\lim_{n \rightarrow \infty} x_n = 0$. Indeed, if $x_n \rightarrow 0$, then the partial sums form a Cauchy sequence since

$$|s_N - s_M| = |x_{N+1} + \dots + x_M| \leq \max_{N < n \leq M} |x_n|$$

and conversely if s_N converge, then $s_N - s_{N-1} = x_N \rightarrow 0$.

Proposition 6.6. *Let K be a field and let $\nu : K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation. Let $A \subset R_\nu$ be a set of representatives of $k_\nu = R_\nu/m_\nu$. Then every element of $\hat{R}_\nu$ can be written uniquely as $\sum_{m=0}^{\infty} a_m t^m$ with $a_n \in A$ where a representative Cauchy sequence $(x_n) \geq 0$ is given by the partial sums $x_n = \sum_{m=0}^n a_m t^m$.*

Proof. Let $x \in \hat{R}_\nu$ given. There is exactly one $a_0 \in A$ such that $|x - a_0|_\nu < 1$. Then, we let $x = a_0 + tb_1$ for some $b_1 \in \hat{R}_\nu$. Then, there is exactly one $a_1 \in A$ such that $|b_1 - a_1|_\nu < 1$, and then $b_1 = a_1 + tb_2$ for some $b_2 \in \hat{R}_\nu$. In this way, we get for every n , $x = a_0 + a_1 t + a_2 t^2 + \dots + a_n t^n + b_{n+1} t^{n+1}$ with $a_i \in A$ and $b_{n+1} \in \hat{R}_\nu$. Now, $b_{n+1} t^{n+1} \rightarrow 0$, hence $x = \sum_{m=0}^{\infty} a_m t^m$. $\square$

Thus, given two elements $x = \sum_{m=0}^{\infty} a_m t^m, y = \sum_{m=0}^{\infty} b_m t^m$, using the addition and multiplication in R_ν , we can write

$$\begin{aligned} \sum_{m=0}^{\infty} a_m t^m + \sum_{m=0}^{\infty} b_m t^m &= \sum_{m=0}^{\infty} c_m t^m \\ \left(\sum_{m=0}^{\infty} a_m t^m \right) \cdot \left(\sum_{m=0}^{\infty} b_m t^m \right) &= \sum_{m=0}^{\infty} d_m t^m \end{aligned}$$

However, in general, determining $(c_m)_{m \geq 0}$ and $(d_m)_{m \geq 0}$ from $(a_m)_{m \geq 0}$ and $(b_m)_{m \geq 0}$ can be quite complicated. In particular, the above proposition does not mean that every $\hat{R}_\nu$ is isomorphic to $k_\nu[[t]]$. On the other hand, if we want to compute c_m and d_m for all $m \leq n$ for some fixed n , we can do this in R_ν by working modulo t^{n+1} , thanks to the following isomorphism of rings:

$$R_\nu/t^{n+1}R_\nu \simeq \hat{R}_\nu/t^{n+1}\hat{R}_\nu \quad \text{for all } n \geq 0.$$

Example 6.7. *As in Example 5.3, let p be prime number and consider the discrete valuation ring $\mathbb{Z}_{(p)}$ with the valuation $\nu_p : \mathbb{Q} \rightarrow \mathbb{Z} \cup \{\infty\}$. The field of p -adic numbers is*

$$\mathbb{Q}_p := \hat{\mathbb{Q}}_{\nu_p}$$

and we define p -adic integers as

$$\mathbb{Z}_p := \{a \in \mathbb{Q}_p : \nu_p(a) \geq 0\}$$

which is a local ring with maximal ideal (p) , and $\mathbb{Z}_p/(p) \simeq \mathbb{Z}/p\mathbb{Z}$.

Exercise 6.8. Show that $\mathbb{Z}[[x]]/(x-p)$ is isomorphic to $\mathbb{Z}_p$.

Theorem 6.9 (Hensel's lemma). Let K be a field and let $\nu : K \rightarrow \mathbb{Z} \cup \{\infty\}$ be a discrete valuation. Let $f(X) = \sum_{i=0}^n a_i X^i \in R_\nu[X]$ with $a_n \in R_\nu \setminus \mathfrak{m}_\nu$. Denote by

$$\bar{} : R_\nu \rightarrow R_\nu/\mathfrak{m}_\nu = k_\nu$$

the reduction homomorphism to the residue field, and $\bar{f}(X) = \sum_{i=0}^n \bar{a}_i X^i \in k_\nu[X]$. Suppose that there exists $x_0 \in R_\nu$ such that

$$(4) \quad \bar{f}(\bar{x}_0) = 0 \quad \text{and} \quad \bar{f}'(\bar{x}_0) \neq 0$$

where $f'(X) := \sum_{i=1}^n i a_i X^{i-1}$ is the (formal) derivative of $f(X)$. Then, there exists a unique root x of f in $\hat{K}_\nu$, satisfying $|x - x_0| < |f'(x_0)|$.

Furthermore, we have

$$|x - x_0| = \frac{|f(x_0)|}{|f'(x_0)|} < |f'(x_0)| \quad \text{and} \quad |f'(x)| = |f'(x_0)|.$$

Proof. By (4), we have that $f(x_0) \in \mathfrak{m}_\nu$ and $f'(x_0) \notin \mathfrak{m}_\nu$. Thus, $|f(x_0)| < 1$ and $|f'(x_0)| = 1$. In particular, we have

$$(5) \quad |f(x_0)| < |f'(x_0)|^2$$

and in fact this is the only condition we will use in the rest of the proof, so the assumption (4) can be replaced with the weaker assumption (5). Since $\left| \frac{f(x_0)}{f'(x_0)} \right| < |f'(x_0)| \leq 1$, there exists $y_0 \in \mathfrak{m}_\nu$ such that $f(x_0) + y_0 f'(x_0) = 0$. Let us define polynomials $f_1, f_2, \dots \in R_\nu[X]$ via the identity

$$f(X + Y) = f(X) + f_1(X)Y + f_2(X)Y^2 + \dots$$

Then $f_1(X) = f'(X)$. We have

$$|f(x_0 + y_0)| \leq \max_{j \geq 2} |f_j(x_0) y_0^j|.$$

Since $f_j(X) \in R_\nu[X]$ and $y_0 \in R_\nu$, we have $|f_j(x_0)| \leq 1$. Hence,

$$|f(x_0 + y_0)| \leq |y_0|^2 = \frac{|f(x_0)|^2}{|f'(x_0)|^2} < |f(x_0)|.$$

Similarly,

$$|f'(x_0 + y_0) - f'(x_0)| \leq |y_0| < |f'(x_0)|$$

and so $|f'(x_0 + y_0)| = |f'(x_0)|$ by the non-archimedean property of the norm. We now put $x_1 = x_0 + y_0$ and note that, similarly as in (5), we have

$$|f(x_1)| < |f'(x_1)|^2,$$

so we can repeat this process. In this way, we obtain a sequence $(x_n)_{n \geq 0}$ using Newton's formula $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ with the properties:

- (1) $|x_n| \leq 1$ hence $x_n \in R_\nu$.
- (2) $|f'(x_n)| = |f'(x_0)|$
- (3) $|f(x_{n+1})| \leq \frac{|f(x_n)|^2}{|f'(x_n)|^2} = \frac{|f(x_n)|^2}{|f'(x_0)|^2} < |f(x_n)|$ and so $f(x_n) \rightarrow 0$ as $n \rightarrow \infty$.
- (4) $|x_{n+1} - x_n| = |y_n| = \frac{|f(x_n)|}{|f'(x_n)|} = \frac{|f(x_n)|}{|f'(x_0)|} \rightarrow 0$ as $n \rightarrow \infty$.

It follows that $(x_n)_{n \geq 0}$ is a Cauchy sequence with respect to ν . Then, the element $x := \lim_{n \rightarrow \infty} x_n \in \hat{K}_\nu$ exists and $f(x) = 0$. To see that $|x - x_0| = |f(x_0)| / |f'(x_0)|$, let us observe that

$$|x_{n+1} - x_n| = \frac{|f(x_n)|}{|f'(x_n)|} < \frac{|f(x_0)|}{|f'(x_0)|}$$

for all $n \geq 1$. Now, note that $|x_1 - x_0| = |f(x_0)| / |f'(x_0)|$ and if $|x_n - x_0| = |f(x_0)| / |f'(x_0)|$ for some n , then it follows that $|x_{n+1} - x_0| = |f(x_0)| / |f'(x_0)|$ since we have $|x_{n+1} - x_0| = |x_n - x_0|$ by Lemma 6.1.

Finally to see the uniqueness, suppose $\bar{x}$ is another solution with $|\bar{x} - x_0| < |f'(x_0)|$ and $x \neq \bar{x}$. Then, let $\tilde{x} = x + \tilde{y}$. We have

$$0 = f(x + \tilde{y}) - f(x) = f'(x)\tilde{y} + f_2(x)\tilde{y}^2 + \dots$$

But,

$$|\tilde{y}| = |x - x_0 + x_0 - \bar{x}| \leq \max\{|x - x_0|, |\bar{x} - x_0|\} < |f'(x_0)| = |f'(x)|$$

and $|f_i(x)| \leq 1$, the first term on the right side of the above equation has strictly greater norm than the others, hence the sum cannot be 0, which is a contradiction. $\square$

Corollary 6.10. *Let*

$$f(X) = X^{p-1} - 1 \in \mathbb{Z}[X] \subset \mathbb{Z}_p[X].$$

Then, for every $0 \neq \bar{\xi} \in \mathbb{Z}/p\mathbb{Z}$ there exists a unique $\xi \in \mathbb{Z}_p$ such that $f(\xi) = 0$ and $[\xi] = \bar{\xi} \in \mathbb{Z}_p/(p) = \mathbb{Z}/p\mathbb{Z}$.

Proof. Let $0 \neq \bar{\xi} \in \mathbb{Z}/p\mathbb{Z}$ and let ξ_0 be a lift of $\bar{\xi}$ in $\mathbb{Z}_p$. Since

$$\bar{f}(X) = X^{p-1} - 1 = \prod_{0 \neq x \in \mathbb{Z}/p\mathbb{Z}} (X - x),$$

we have that

$$\bar{f}(\bar{\xi}) = 0 \quad \text{and} \quad \bar{f}'(\bar{\xi}) = (p-1)\bar{\xi}^{p-2} \neq 0 \quad \text{in } \mathbb{Z}/p\mathbb{Z}.$$

Hence, by Theorem 6.9, f has a root ξ in $\mathbb{Z}_p$ such that $|\xi - \xi_0| < 1$ and therefore, ξ is also a lift of $\bar{\xi}$. $\square$

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